Espíndola's objections to *A CH obstruction to a prescribed categoricity threshold*

An AI-only investigation, with a proofreading of Theorem 4.1 of arXiv:2301.13167

25 October 2026

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About this document

What it is. The community repository mathcommunity/oai-math-revised collects notes on the 719 manuscripts that OpenAI released on 6 October 2026. One of them, *A CH obstruction to a prescribed categoricity threshold* (below: *the manuscript*), states as Theorem 1.1 that under CH there is an abstract elementary class with Löwenheim–Skolem number $\aleph_0$ that has two non-isomorphic models of size $\beth_{\omega_2}$ but is categorical in every cardinal from $\Lambda = \beth_{(2^{\aleph_1})^+}$ on. This conflicts with Theorem 4.1 of Christian Espíndola's preprint arXiv:2301.13167. On 17 October 2026 Espíndola objected publicly on X (thread in Appendix A): he wrote that the manuscript misuses a cardinal bound (objection E1) and that its Proposition 7.2 does not refute his proof (objection E2).

The repository's maintainers asked AI systems to look into this, in order to obtain independent information that domain experts could use. **This document records that investigation. It is not a verdict and has not been checked by any person.** All mathematical conclusions in it are AI-generated and may be wrong. They are meant to point experts to the places worth checking.

How it was done.

- *Round 1* (*the objections*). Three instances of OpenAI's GPT-6 Astra (via Codex) investigated, separately, objection E1, the Lean formalization, and objection E2. An instance of Anthropic's Claude Opus 5.5 answered all three questions independently. A further Astra instance acted as judge: it read the four reports, checked their main claims against the sources, and wrote the report in Part 1.
- *Lean check.* The manuscript's Theorem 1.1 has a Lean formalization in OpenAI's release (statement `CHObstruction.lean`, proof module `OAI.ModelTheory.Categoricity.Main`). Lean's official comparator was run on it in a fresh directory and accepted it (Appendix B).
- *Round 2* (*proofreading Espíndola's Theorem 4.1*). One Astra instance refereed the proof of Theorem 4.1 seeing only Espíndola's two preprints, without being told about the manuscript. Another Astra instance and a Claude Opus instance walked the proof through the manuscript's concrete class. An Astra judge wrote the addendum in Part 2.

Sources and versions. The manuscript and its Lean files at repository commit `b36659d` (links in this document point to that commit, with line anchors); Christian Espíndola, *A complete classification of categoricity spectra of accessible categories with directed colimits*, arXiv:2301.13167v1, and *A proof of Shelah's eventual categoricity conjecture and an extension to accessible categories with directed colimits*, arXiv:1906.09169v7 (references of the form `thresholds.tex:359` or `scc3.tex:617` are to lines of their arXiv source files); Lean 4.34.1 with Mathlib as pinned by the release.

Numbering. Theorem numbers follow the manuscript's PDF. Its TeX file names do not match its section numbers: `04-endpoint.tex` is Section 4 and `03-categoricity.tex` is Section 5. The round-1 briefs (Appendix C) stated this mapping wrongly; the investigators noticed and used the PDF numbering.

How to read it. Parts 1 and 2 are the judges' reports, about 2,700 words together. The appendices reproduce the briefs and the eight underlying reports unchanged, except that local file paths were replaced by public links. They overlap and occasionally disagree; the judges' reports discuss the disagreements.

Contents. Part 1: report on the objections. Part 2: addendum on the proof of Theorem 4.1. Appendix A: the X thread. Appendix B: the Lean check. Appendix C: the briefs. Appendix D: round-1 reports. Appendix E: round-2 reports.

Part 1. Report on the objections

1. Summary

Here, "categorical" means that models of a specified size exist and are all isomorphic. In the supplied manuscript, Proposition 5.7 states a different bound from the one attributed to it in E1, and the disputed endpoint models come from a separate construction. Direct source checks and the reported Lean comparator result provide strong evidence for the manuscript's Theorem 1.1 under CH; no human has checked this investigation's mathematical conclusions. The inspected Lean statement matches that theorem's existential conclusion, with no identified vacuity or weakened AEC axiom. Theorem 1.1 and Espíndola's Theorem 4.1 have incompatible conclusions under CH. Proposition 7.2 establishes a more specific obstruction on the original category; its applicability to the modified category mentioned in the reply remains unresolved.

2. Objection E1

The three relevant cardinals have different roles:

Cardinal	Role
$(\beth_{\omega_1})^+ = \lvert V_{\omega_1}\rvert^+$	Auxiliary set size in Lemma 5.2's contradiction
$\beth_{\omega_2}$	Size of the two endpoint models
$\Lambda = \beth_{(2^{\aleph_1})^+}$	Proposition 5.7's bound and the categoricity-tail threshold

Proposition 5.7 actually concludes

$$B^M \text{ uncountable} \quad \Longrightarrow \quad |I^M| < \Lambda.$$

It does **not** conclude $|I^M| < \beth_{\omega_2}$. See PDF p. 24, 03-categoricity.tex:440, with Λ defined at line 7.

Lemma 5.2 prohibits one fixed color per arity passing every finite test. Its proof copies the hypothetical compatible diagrams onto a newly chosen set X of size $|V_{\omega_1}|^+$, then obtains an impossible injection into V_{ω_1}. Nothing requires X to be a subset of the original model. The smaller cardinal is therefore an auxiliary contradiction device, not a bound asserted for every model. See PDF pp. 19–21, 03-categoricity.tex:76, line 93.

The bridge to Proposition 5.7 is a separate partition argument. Whole color profiles have at most $\xi = 2^{\aleph_1}$ possibilities; $\mathrm{cf}(\Lambda) = \xi^+ > \xi$ permits selection of fixed profiles with finite witnesses. One base coordinate avoids their countably many exceptional sets, producing Lemma 5.2's prohibited pattern. I reconstructed this argument, including the finite partition estimate. See PDF pp. 21–24, 03-categoricity.tex:228, line 398, line 445.

The endpoint existence claim is independent of those two results. Theorem 4.5 constructs models with index sort V_{ω_2}, and bases respectively of size $\aleph_1$ and $\aleph_0$. Both total sizes are $\beth_{\omega_2}$; the named base predicate prevents an isomorphism. The final proof explicitly invokes Theorem 4.5 for this conclusion. See PDF p. 18, 04-endpoint.tex:335, and p. 26, 05-conclusion.tex:3.

Confidence: high in the statement and dependency findings; medium-high in the reconstructed mathematical arguments. No E1 gap was identified in this version.

3. The Lean formalization

`main` asserts, assuming CH, the existence of a countable finitary relational language and an AEC K satisfying:

- $\mathrm{LS}(K) = \aleph_0$;
- $\beth_{\omega_2} = h(\aleph_0)$, where $h(\kappa) = \beth_{(2^\kappa)^+}$;
- two nonisomorphic models of size $\beth_{\omega_2}$;
- categoricity, including existence, at every $\mu \geq \Lambda$.

These match Theorem 1.1, PDF p. 2, 00-introduction.tex:87. Programmatic comparison confirmed identical public definition blocks and `main` types in the challenge and solution sources.

I checked the principal semantic failure modes. Isomorphisms are relation-preserving and relation-reflecting bijections; categoricity asserts existence; LS includes arbitrary requested subsets and language size; AEC axioms include isomorphism invariance, coherence, unrestricted nonempty ordinal unions, and smoothness. Carriers are arbitrary universe-relative `ZFSets`. See Model.lean:26, line 58.

The endpoint proof, `EndpointModels.twoModels`, uses the explicit V_{ω_2} construction and tuple-dependent countable exceptions. Its inspected proof path avoids `Tests.no_pattern` and `TailBound.index_lt`, the counterparts of Lemma 5.2 and Proposition 5.7. Their presence in imported files does not make them dependencies of this argument.

lean-check.txt reports successful comparator acceptance using Lean's default kernel, with only `propext`, `Quot.sound`, and `Classical.choice` permitted. This is stronger evidence than compilation alone. [*Update after this report was written: the comparator was rerun in a fresh directory in which the solution had never been built, as the comparator's trust model asks; it again accepted the solution with the default kernel and only the three standard axioms. The second kernel (nanoda) was not used.*]

Acceptance establishes the encoded proposition, subject to the checking infrastructure. It does not certify every prose argument, establish a formal translation into ZFC, or verify Proposition 7.2. Given the inspected definitions, it bears directly on E1's disputed endpoint. **Confidence: high in statement correspondence; the kernel check is reported above.**

4. Objection E2 and the relation between the theorems

Espíndola's Theorem 4.1 states that **any AEC** categorical at some $\lambda \geq \beth_{(2^{\mathrm{LS}(K)})^+}$ is categorical at every cardinal above or equal to that threshold. Its statement imposes no amalgamation, joint embedding, tameness, or no-maximal-models hypothesis. See his PDF p. 6, arXiv:2301.13167v1, thresholds.tex:354.

The inspected class fits that scope: its LS number is $\aleph_0$, and its strong embeddings provide concrete directed colimits. The internal-size convention does not supply an escape here: the manuscript verifies agreement with underlying cardinality, including countable models. See manuscript PDF p. 29, 06-booleanization.tex:108.

Under CH,

$$\beth_{(2^{\aleph_0})^+} = \beth_{\omega_2} < \Lambda.$$

Applying Theorem 4.1 at $\lambda = \Lambda^+$ would force categoricity at $\beth_{\omega_2}$, where Theorem 1.1 supplies two models. **This statement-level conflict is high-confidence and does not depend on Proposition 7.2.**

Assuming ZFC is consistent, both claims cannot be ZFC theorems: together they would prove ¬CH, contrary to CH's relative consistency. This does not locate a particular proof error.

Proposition 7.2 concerns the canonical point associated with a model H, on the category of countable models with the **original strong embeddings**. Its argument checks out as follows: choose countable $A \to H$ covering all of H's countable base. The arrows from A that add a base point form a dense cosieve, because every countable model permits such an extension. But $A \to H$ cannot factor through any such arrow: injectivity would force a new base point to collide with an existing one. Therefore the canonical point does not factor through the full double-negation subtopos. See manuscript PDF pp. 28–30, 06-booleanization.tex:15, line 153.

Espíndola's written proof asserts that the large categorical model supplies a point of that subtopos—his PDF p. 7, arXiv:2301.13167v1, thresholds.tex:359. Proposition 7.2 obstructs the canonical-point reading on the original category. A genuine change of morphisms can change density and points; an equivalent presentation alone transports the obstruction.

The reply does not specify enough to test the modified construction. Preservation of the spectrum would leave the theorem-level conflict intact, while preservation of the LS number and relevant sizes also needs justification. **Confidence: medium-high for the original-site obstruction; unresolved for the intended modified site.**

5. Disagreements, uncertainties, and additional checks

The reports substantially agree on E1 and the theorem-level conflict. Their confidence differs most over whether Proposition 7.2 identifies Espíndola's intended proof step. The narrower formulation above is warranted. The manuscript version underlying the thread also remains unidentified.

I specifically challenged the following possible shared blind spots:

- **Endpoint construction:** The recursion defining $f_\alpha : \omega_1 \to \omega_1$, for $\alpha < \omega_2$, takes only countable suprema. Each countable membership diagram therefore has countably many exceptional coordinates. Monotone closures and fixed enumerations ensure exact compatibility under restriction. No common good coordinate for all tuples is asserted. I found no failure here. PDF pp. 15–17, 04-endpoint.tex:27, line 125, line 236.
- **AEC axioms:** At a directed union, representatives for a fixed tuple are chosen inside one stage, avoiding an unjustified assumption that countably many preselected elements lie together. The LS construction collects exceptional sets for at most $|A| + \aleph_0$ tuples. These checks address substantive risks; neither revealed a gap. PDF pp. 12–14, 02-aec.tex:466, line 582.
- **CH:** Opus's "only in coding" description is incomplete: identifying the prescribed threshold uses CH. Conversely, the coding estimate itself follows without CH from $\aleph_1^{\aleph_0} \leq (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$; the profile bound similarly needs no CH. This observation does not establish an unconditional obstruction at $h(\aleph_0)$.
- **Arithmetic:** One Opus table makes $(\beth_{\omega_1})^+ < \beth_{\omega_1+1}$ strict. Only $\leq$ is guaranteed. The required strict inequality with $\beth_{\omega_2}$ remains valid.

Opus also proposes that a strong extension adding a base point forces the old index size to be at most $\beth_{\omega_1}$. I reconstructed its gluing-and-collapse argument and found it persuasive, though unformalized here. It bounds **extendibility**, not existing uncountable-base models. Its implications for Espíndola's proposed covering families need care: absence from a generating family alone does not prove absence from the generated topology.

6. Questions for domain experts

1. Does the AEC verification handle all standard requirements, especially smoothness at countable cofinality and LS for arbitrary subsets?
2. Is any compatibility or validity condition missing from the endpoint diagrams, or any selection step invalid in Lemmas 5.5–5.6?
3. Precisely which modified category, morphisms, and point are intended at `thresholds.tex:359`, and how are LS number and sizes preserved?
4. In that category, is Proposition 7.2's cosieve available and dense? Does the proposed topology actually equal double negation?

7. Method and limitations

This is an AI-only investigation: three supplied Codex/gpt-6-astra reports, one independent Claude Opus report, and this Codex review. No human has checked these mathematical conclusions.

I read the context, thread, comparator report, and all four investigations; checked PDF numbering with `pdftotext`; reconstructed the central arguments; compared Lean definitions and theorem types programmatically; and scanned proof modules for proof holes and extra axiom declarations. I made no edits or network calls and ran no Lean build or comparator. This was not a complete audit of Mathlib, the checking infrastructure, or either topos-theoretic development.

Part 2. Addendum: proofreading of Theorem 4.1 of arXiv:2301.13167

1. Summary

The principal concern is the opening proof paragraph at `thresholds.tex:359`, PDF p. 7. The blind referee independently questions both the proposed generators of the dense topology and the passage from the categorical model to a point of the double-negation subtopos—the same passages challenged concretely by both informed reports. Direct checking supports the canonical-point obstruction with high confidence on the original category of strong embeddings. The stronger objections concerning generators and closedness are persuasive, with moderate confidence in the reconstructed categorical arguments. Changing morphisms remains a possible change of construction, but preservation of the categoricity spectrum alone does not preserve the LS-dependent threshold. Nothing found here resolves the statement-level conflict recorded in the first-round report.

2. Principal steps flagged

All three quotations below occur at arXiv:2301.13167v1, thresholds.tex:359, PDF p. 7.

The categorical-model point. The proof states:

> "Clearly the model of size λ must be one of its points"

The conventions check matters. Put $C = K_{\aleph_0}$, with strong embeddings. Set^C means presheaves on C^{op}; covering sieves therefore correspond to collections of arrows **out of** a model in C. This agrees explicitly with arXiv:1906.09169v7, scc3.tex:670, PDF p. 20. A point is a geometric morphism from Set, not an element of a model.

Let H be the CH class's categorical model of size $\lambda = \Lambda^+$, where $\Lambda = \beth_{(2^{\aleph_1})^+}$. Its base is countable. Its canonical presheaf point satisfies

$$p_H^*(\mathrm{Hom}(A, -)) = \mathrm{Hom}(A, H)$$

and even preserves countable limits, so the distinction between ordinary points and $\aleph_1$-points does not remove this obstruction.

Choose countable $a : A \to H$ covering the entire base of H. Let R_A comprise the countable extensions of A adding a base point. Every countable extension admits a further such extension, making R_A dense. But a cannot factor through any member of R_A: injectivity would send the additional base point into an already exhausted base. Thus p_H does not respect this dense cover.

I checked this argument and the extension construction in 06-booleanization.tex:15 and line 153, PDF pp. 28–30. **Both informed reports give this obstruction; the blind report independently identifies the missing covering-lifting argument but labels the claim unresolved.** This concerns the canonical point; it does not establish that the Booleanization has no points.

The preceding generator claim. The proof says:

> "the dense topology is generated by those covers $\{M \to N_i\}$"

subject to:

> "any $M \to N$, for an arbitrarily large model N, can be amalgamated with one $M \to N_j$."

Both informed reports strengthen the first-round objection from absence among generators to absence from the **generated topology**. Their additional ingredient checks out:

If $M \leq_K N$ adds a base point, then

$$|I^M| \leq |V_{\omega_1}| = \beth_{\omega_1}.$$

At the new point, all old tuples pass their tests without exceptions. Their compatible diagrams glue to an extensional, well-founded relation whose collapse injects I^M into V_{ω_1}. This uses 02-aec.tex:286 and the gluing argument at 03-categoricity.tex:103, PDF pp. 20–21. It bounds **extensions adding base points**, not existing uncountable-base models.

Consequently, extensions of H retain its base. The matrix construction at 03-categoricity.tex:505, PDF pp. 24–26, allows countable pieces of an amalgam to embed back into H, fixing the required countable submodel. On the universal reading of "arbitrarily large," this makes p_H respect every proposed generator, hence their generated topology, while still rejecting R_A. One can first enlarge H arbitrarily, so the argument need not assume that the phrase includes H itself.

Moderate confidence: the local embedding argument and quantifier interpretation deserve expert scrutiny. **Independent agreement:** the blind report also doubts equality with the dense topology, without using K. Restricting N to small models would change the generator issue but leave the canonical-point obstruction.

Closedness without amalgamation. The proof states:

> "we do not need to use amalgamation, since κ^+-closed models are precisely the κ^+-points of the double negation subtopos"

The earlier paper defines closedness through purity of maps **out of the whole model** (arXiv:1906.09169v7, scc3.tex:617, p. 18). Its Theorem 7.1 identifies **saturated models** with dense-topology points and explicitly begins:

> "Assume *GCH* and amalgamation."

Its proof uses amalgamation to identify the dense and atomic topologies (arXiv:1906.09169v7, scc3.tex:688, p. 20); the closedness-to-saturation passage also explicitly invokes amalgamation at line 650 (arXiv:1906.09169v7, scc3.tex:650), p. 19.

Rigidity and the local embedding argument support both informed reports' claim that H is $\aleph_1$-closed, despite failing the required saturation and point conditions. This reconstruction does not depend on the earlier categoricity-implies-closedness lemma. The blind report independently flags the missing amalgamation justification, but supplies no concrete closed counterexample.

3. Other doubtful steps, ranked

1. **Changing morphisms inside slices.** The blind report identifies a substantive issue at arXiv:2301.13167v1, thresholds.tex:281–294, p. 3. Restricting $\mathrm{Hom}_C(M,-)$ to Boolean arrows retains all original arrows out of M; it does not become $\mathrm{Hom}_{C^B}(M,-)$. The asserted pullback/slice identification needs justification. The informed reports do not independently audit this calculation.
2. **Applying the transfer machinery.** Lemma 2.1 requires amalgamation and stability. Theorem 3.2 requires an initial categoricity interval, not merely two cardinals. Applying Theorem 3.1 at κ^+ gives 2^{κ^+}, not the asserted 2^{κ}. See arXiv:2301.13167v1, thresholds.tex:277, 329, 339, PDF pp. 3, 5–6. Blind and informed Astra flag these discrepancies.
3. **The numerical cutoff.** The final Morley-method invocation does not supply its required coding argument or track LS through the modifications. The blind report independently flags this.

Two qualifications prevent overcounting objections. The hooked Hom-arrows could mean their **images**, repairing that notation. Also, conditional on a genuine Boolean-topos point, its subterminal truth filter selects an atom; the blind report correctly offers this repair. Astra supports countable instability by distinct colors; Opus leaves stability unresolved. Neither conclusion is needed for the principal obstruction.

4. Possible ways out

LS versus accessibility. K has LS number $\aleph_0$ and is $\aleph_1$-accessible. Using the latter changes the beth bound to Λ. This matters for the accessibility-index formulation of Theorem 6.1, whose strict cutoff needs separate assessment. It does not reinterpret Theorem 4.1 (arXiv:2301.13167v1, thresholds.tex:355), p. 6, which explicitly uses $LS(K)$. Internal and underlying sizes agree here.

Modified morphisms. An equivalent presentation transports the obstruction. A genuine restriction can remove it: requiring embeddings to preserve the entire base raises LS to $\aleph_1$, changing the threshold to Λ. This is an illustrative modification, not one specified in the reply.

Opus's stronger argument is persuasive **within its stated scope**: the same structures, ordinary injective embeddings, coherence, and LS = $\aleph_0$. Countable-submodel existence and isomorphism invariance then recover dense base-adding extensions. It does not rule out every unspecified categorical replacement. The reply still needs explicit arrows, a small-object site, and preservation of the relevant LS number and sizes.

5. Questions for experts

- What exactly are the arrows and quantifiers in the proposed generating families?
- Does rigidity plus the local embedding argument establish continuity for every generator?
- What replaces amalgamation in the closedness-to-point assertion?
- How does the intended modification preserve the LS-dependent bound?

6. Method and limitations

I read all three reports, checked source passages and PDF pagination, verified that the blind source copies match, and reconstructed the central arguments. I also checked the 2023 and earlier arXiv records. I did not rerun Lean or audit every cited theorem. The comparator result does not certify these topos arguments. **This investigation is AI-only; none of its mathematical conclusions has been checked by a person. Topos-theoretic reasoning is where this AI reading is least reliable.**

Appendix A. The X thread

As copied from x.com by a maintainer on 25 October 2026.

```
Artem Chernikov (@archernikov), Oct 7: Some results from the Open AI slopacalypse of
 ↪  interest to mathematical logicians: [...]
  Shelah's eventual categoricity conjecture for AEC's holds [Eventual-Categoricity
 ↪  paper] but not starting with the expected bound [CH obstruction paper]

Jason Rute (@JasonRute), Oct 8: Do you know the status of Espindola's claimed proof
 ↪  the eventual categoricity conjecture?
Artem Chernikov, Oct 8: My understanding is that it is certainly not currently
 ↪  accepted by the (rather small) AEC community.
Jason Rute, Oct 9: We will see if this claimed proof is accepted faster. (I think
 ↪  there is no Lean for the main result, just the obstruction, right?)

Christian Espindola (@christianrubene), Oct 17: Hello, two comments:
1) there are no topos theorists experts in the AEC group, there are however two
 ↪  people that understand topos theory and that have agreed to serve as external
 ↪  reviewers for my habilitation thesis. We are in the process of formalizing my
 ↪  papers in Lean.

Christian Espindola, Oct 17
 ↪  (https://x.com/christianrubene/status/2111425004667572321):
2) The Open AI obstruction paper contradicts one of my results in the explicit
 ↪  thresholds paper, especifically their Proposition 7.2 is saying that models of a
 ↪  categoricity cardinal k, which are k-closed, do not factor through the double
 ↪  negation subtopos. I have identified a couple of errors in the Open AI preprint
 ↪  that should invalidate their claims even with a Lean certification. They claim
 ↪  that their AEC hs in particular two sorts, B and I, with B at most of
 ↪  cardinality \aleph_1, and that the size of an infinite model M is max(|I|^M,
 ↪  |B|^M), and they prove in Proposition 5.7 that if B^M is uncountable, then I^M
 ↪  is less than \beth_(\delta^+) with \delta=\omega_1. This particular proof though
 ↪  seems to be based on their Lemma 5.2, where they clearly used the bound
 ↪  |V_\delta|^+ corresponding to the smaller (\beth_\delta)^+). This inaccuracy is
 ↪  what allows them to have two non-isomorphic models M, N one with countable B^M
 ↪  and another with uncountable B^N. But this cannot occur at \beth_(\delta^+), by
 ↪  their own argument, only at (\beth_\delta)^+). This wouldn't contradict the
 ↪  prescribed threshold.
```

```
As for Proposition 7.2, they try to falsify my Theorem 4.1 in a way that does not
 ↪  work: I use an axiomatization of a category of models categorically equivalent
 ↪  to a modified AEC (same models, different morphisms) which do not affect the
 ↪  categoricity spectrum. END

Jason Rute, Oct 17: Thanks for the follow up! As for Lean, it is indeed useful for
 ↪  contested results (like @cameronfreer et al's Lean proof of the Vaught
 ↪  Conjecture counterexample). Are you using AI to help write the Lean?
Jason Rute, Oct 17: As for the disagreement between your and the OpenAI's results,
 ↪  how do you reconcile the proven Lean result in CHobstruction.lean? Is the Lean
 ↪  statement misformalized or not relevant to the above? (Note the natural language
 ↪  proof could have mistakes even if the Lean is correct.)
Christian Espindola, Oct 17: I have not looked at the Lean statement, maybe it
 ↪  compiles well, but I think it is formalizing something else. Lemma 5.2 uses one
 ↪  threshold, Proposition 5.7 which cites it uses another which is syntactically
 ↪  similar (one interchange of successor and \beth operator). Strange.
```

Appendix B. The Lean check

```
Lean check (run 2026-10-25 on a university server, Lean v4.34.1, Mathlib d13f23b7,
 ↪  lean4export and comparator built against v4.34.1, landrun sandbox):
`lake env comparator ComparatorChallenges/CHObstruction.json` (challenge
 ↪  ComparatorChallenges.CHObstruction, solution OAI.ModelTheory.Categoricity.Main,
 ↪  theorem OAI.CHObstruction.main, permitted axioms propext, Quot.sound,
 ↪  Classical.choice) printed:
  "Running Lean default kernel on solution. / Lean default kernel accepts the
 ↪  solution / Your solution is okay!"
Caveats: a first comparator attempt (failed at export because of a tool-version
 ↪  mismatch) and a `lake build` of the solution ran in the same directory before
 ↪  the successful comparator run, so this is not the comparator's strict "never
 ↪  compiled the solution before" setting; nanoda was not used. arXiv lists only v1
 ↪  of 2301.13167.
```

Second, fresh run (no compiled solution present before the comparator ran): the comparator printed "Lean default kernel accepts the solution" and "Your solution is okay!". The second kernel (nanoda) was not used.

Appendix C. Briefs

Round 1: CONTEXT.md

Independent investigation: Espíndola's objections to "A CH obstruction to a prescribed categoricity threshold" Purpose: gather independent, carefully argued information on a public disagreement, for maintainers of a community repository who may pass the results to domain experts. We do NOT adjudicate; we want precise, checkable findings, honest uncertainty, and the exact places a human expert should look. Never edit the source files. No network.

Materials (all under): - paper/ : the OpenAI manuscript (paper.pdf; TeX in paper/build/, sections

00-introduction ... 06-booleanization; file 00 is Section 1, so 04-endpoint.tex is Section 5 and 06-booleanization.tex is Section 7). Use the PDF (pdftotext) to confirm numbering such as "Lemma 5.2", "Proposition 5.7", "Proposition 7.2", "Theorem 1.1". - x-thread.txt : the X thread, including Christian Espíndola's objections (verbatim). - espindola-src/2301.13167/ : Espíndola, "A complete classification of categoricity spectra of accessible categories with directed colimits" (arXiv v1, TeX in x/thresholds.tex, paper.pdf) — his "explicit thresholds paper", Theorem 4.1. - espindola-src/1906.09169/ : Espíndola's earlier claimed proof of eventual categoricity (v7). - lean/ : OpenAI's Lean formalization: CHObstruction.lean (challenge statement, `sorry` by design), CHObstruction.json (names the proof module OAI.ModelTheory.Categoricity.Main), 240.md (OpenAI's scope note), and lean/OAI/... the 15 proof modules (imports only Mathlib). A separate machine is running Lean's official comparator on it; its result will be given to the final judge.

Espíndola's two objections (paraphrase; read x-thread.txt for his words): (E1) Proposition 5.7 (if B^M is uncountable then |I^M| < beth_{delta^+}, delta = omega_1) relies on Lemma 5.2, which only gives the bound |V_delta|^+ = (beth_delta)^+. So the two non-isomorphic models (countable B vs uncountable B) exist only at (beth_delta)^+, not at beth_{delta^+} = beth_{omega_2}; the prescribed threshold is not contradicted. (E2) Proposition 7.2 does not refute his Theorem 4.1, because he axiomatizes a category equivalent to a modified AEC (same models, different morphisms), which does not change the categoricity spectrum. Jason Rute's point: the Lean statement is proved; is it misformalized or irrelevant to E1? Espíndola: "I have not looked at the Lean statement ... I think it is formalizing something else."

Report format: Markdown, with (1) a short answer, (2) the detailed argument with exact citations (file:line or "Lemma 5.2, p. N"), (3) a confidence level (low/medium/high) for each conclusion and why, (4) what you could not determine, (5) concrete questions a domain expert could answer quickly. Distinguish "the paper's proof has a gap" from "the theorem is false". Quote formulas exactly as written in the sources.

Round 1: Q1.md

Your question (E1): Check Espíndola's objection E1 against the manuscript itself. Identify Lemma 5.2 and Proposition 5.7 and every place the bounds beth_{delta^+}, (beth_delta)^+, |V_delta|^+, H(K), beth_{omega_2} are used. Is there a real mismatch between what Lemma 5.2 proves and what Proposition 5.7 uses? If so, is it a typo that a correct argument repairs, a gap that can be filled, or does it break Theorem 1.1(ii) (two nonisomorphic models of size beth_{omega_2})? Reconstruct the relevant argument yourself in enough detail to decide. Also check Espíndola's description of the class (two sorts B, I, |B| <= aleph_1, model size max(|I^M|, |B^M|)) against the paper.

Round 1: Q2.md

Your question (Lean): Read lean/CHObstruction.lean and lean/240.md, then the proof modules under lean/OAI/. (a) State precisely, in mathematical language, what the Lean theorem OAI.CHObstruction.main asserts, including all definitions it depends on (AEC axioms, strong embeddings, Löwenheim–Skolem number, categoricity, cardinals such as beth_{omega_2}, H(K), Lambda, the CH hypothesis). (b) Is it a faithful formalization of the manuscript's Theorem 1.1? Look for vacuity, weakened definitions, wrong cardinals, or anything that makes it "formalize something else". (c) Locate the Lean counterparts of the manuscript's Lemma 5.2 and Proposition 5.7 and say which cardinal bound the Lean proof actually uses there. (d) Assuming the Lean proof is accepted by the kernel with standard axioms, what does that imply for objection E1?

Round 1: Q3.md

Your question (E2): Read Espíndola's arXiv:2301.13167 (espindola-src/2301.13167/x/thresholds.tex), especially Theorem 4.1 and its proof, and the manuscript's Section 7 (06-booleanization.tex, incl. Proposition 7.2 and "The precise comparison"). (a) Does Proposition 7.2 identify a step that Espíndola's proof actually uses, given his reply that he works with a category equivalent to a modified AEC (same models, different morphisms)? (b) Independently of where his proof might break: do the hypotheses of his Theorem 4.1 cover the AEC of the manuscript's Theorem 1.1 (language, LS number, CH or not, any amalgamation/tameness/arbitrarily-large-models assumptions, which threshold exactly)? If they do, Theorem 1.1 and Theorem 4.1 cannot both be true; say so explicitly, and state precisely which threshold each uses. Compare the threshold in his Theorem 4.1 with beth_{($2^{\{\text{aleph_0}\}}$)+} and with (beth_{omega_1})^+.

Round 1: JUDGE.md

You are the final judge of an independent investigation. Read CONTEXT.md (note: its section mapping is wrong — paper.tex inputs 04-endpoint.tex as Section 4 and 03-categoricity.tex as Section 5; use the PDF numbering), x-thread.txt, lean-check.txt, and the four investigator reports: astra-Q1-report.md, astra-Q2-report.md, astra-Q3-report.md (three Codex/gpt-6-astra investigators) and opus-report.md (an independent Claude Opus investigator who answered all three questions). You may consult the sources (paper/, espindola-src/, lean/) to resolve disagreements or to test the investigators' key claims yourself — do test the load-bearing ones (what Proposition 5.7 concludes; that Theorem 1.1(ii) rests on Theorem 4.5 and not on Lemma 5.2/Prop 5.7; that the Lean `main` statement matches Theorem 1.1 and the Lean two-model proof avoids those lemmas; that Espíndola's Theorem 4.1 as written applies to this class; and the status of Prop 7.2 versus his "modified AEC" reply). Be adversarial toward the investigators' consensus: look for what they could all have missed (e.g. a flaw in the Theorem 4.5 construction or the AEC axioms, an issue with how CH is used, a definition in Lean that trivializes something).

Write the final report (Markdown) for maintainers who are not experts and who may forward it to domain experts in set theory / AEC model theory / topos theory. Structure: 1. Summary (at most 8 sentences, neutral, no verdict language like "Espíndola is wrong"; say what the investigation found and how confident it is). 2. Objection E1: findings, with exact citations (PDF page + TeX file:line), and confidence. 3. The Lean formalization: what it states, what the comparator result does and does not establish, and its bearing on E1. 4. Objection E2 and the logical relation between Theorem 1.1 and Espíndola's Theorem 4.1 (statement-level conflict vs. location of an error in a proof). 5. Points of disagreement or uncertainty among the investigators, and anything you found that they missed. 6. Questions for domain experts (short, concrete, answerable). 7. Method and limitations (AI-only investigation; no human has checked; models used; what was and was not run). Keep it under 2000 words. Neutral, careful, specific. Write it as your final message.

Round 2: BRIEF-blind.md

Proofreading request You are refereeing a mathematics preprint. Read-only; no network; do not read anything outside the folder

Material: - 2301.13167/ : Christian Espíndola, "A complete classification of categoricity spectra of accessible categories with directed colimits" (arXiv v1, 2023). TeX: x/thresholds.tex; PDF: paper.pdf. - 1906.09169/ : Christian Espíndola, "A proof of Shelah's eventual categoricity conjecture and an

extension to accessible categories with directed colimits" (arXiv v7, 2022). TeX: x/scc3.tex; PDF: paper.pdf. The first paper relies on it.

Task: referee the proof of **Theorem 4.1** of 2301.13167 (eventual categoricity for AECs from the threshold beth_{($2^{\{LS(K)\}}$+}) as a careful expert referee would. Go through the proof step by step, including every result it uses from Sections 2–3 of 2301.13167 and from 1906.09169 (trace each citation to its statement and, where load-bearing, its proof). For each step: - state precisely what is claimed, with the objects involved (which category, which morphisms, which site/topology, which point or model, which cardinals); - say whether it follows, with your reasoning; flag unstated hypotheses, changes of category or morphisms, places where a definition is used in two different senses, and cardinal arithmetic that does not add up; - rate it: follows / follows with a fixable gap / unclear / appears wrong. Pay special attention to the topos-theoretic steps (dense / double-negation topology on the presheaf topos of small models, points of subtoposes, Booleanness, the passage to slice toposes, types and isolation) and to where the cardinal threshold enters.

Report (Markdown, under 2500 words): (1) a summary of the proof's architecture; (2) the step-by-step table with citations (file:line and PDF page) and ratings; (3) the most doubtful steps, ranked, each with what an expert should check; (4) what you could not assess. Do not speculate about outside literature or other papers. Write it as your final message.

Round 2: BRIEF-informed.md

Test-case proofreading: Espíndola's Theorem 4.1 against a concrete class Read-only; no network. Background and materials are in (see CONTEXT.md there, but note its section mapping is wrong: paper.tex inputs 04-endpoint.tex as Section 4 and 03-categoricity.tex as Section 5 — use PDF numbering; also read JUDGE-REPORT.md, the result of a first investigation). Espíndola's sources: espindola-src (2301.13167/x/thresholds.tex = the "thresholds" paper with Theorem 4.1; 1906.09169/x/scc3.tex = his earlier paper). The OpenAI manuscript is in paper; its Lean formalization of Theorem 1.1 (two non-isomorphic models at beth_{omega_2} under CH, categoricity from Lambda = beth_{($2^{\{aleph_1\}}$+}) was accepted by Lean's official comparator (default kernel, standard axioms only), and its Lean statement was judged faithful.

Logic: Theorem 4.1 is stated for every AEC. If the OpenAI class K of Theorem 1.1 is an AEC with the stated properties (as the Lean result indicates, modulo faithfulness of the formalization), then Theorem 4.1 fails for K, so its proof must fail for K at some step — or the class is not covered for a reason not yet identified (check this possibility seriously).

Task: 1. Walk through Espíndola's proof of Theorem 4.1 (thresholds.tex around line 354–365 and everything it cites, including Sections 2–3 of that paper and the needed parts of 1906.09169), instantiating each step for the concrete class K (with its morphisms = strong embeddings, LS(K) = aleph_0, CH, the large categorical model of size lambda >= Lambda). 2. Find the FIRST step whose claimed conclusion is false for K, or whose hypotheses K does not satisfy; show this concretely (e.g. with the dense cosieve R_A of the manuscript's Section 7, or other explicit features of K). Also list any other steps that fail for K. 3. Assess Espíndola's reply ("I use an axiomatization of a category of models categorically equivalent to a modified AEC (same models, different morphisms) which do not affect the categoricity spectrum"): is there any choice of morphisms, consistent with what the papers actually write, under which the failing step would go through for K? If such a modification exists, does a later step then fail? 4. Check the alternative: is there any reading under which K is not an AEC in Espíndola's sense (his definition in 1906.09169) or does not meet the hypotheses of Theorem 4.1? Be explicit. Report (Markdown, under 2500 words): short answer; the instantiated

step-by-step walk-through with file:line citations; the failing step(s) with concrete justification and confidence; the assessment of the "modified morphisms" reply; what you could not determine; precise questions for an expert. Write it as your final message.

Round 2: JUDGE2.md

You are the final judge of a second-round proofreading of Christian Espíndola's Theorem 4.1 (arXiv:2301.13167v1, arXiv:2301.13167v1, source file thresholds.tex; his earlier paper 1906.09169 in .../1906.09169/x/scc3.tex). First-round report: JUDGE-REPORT.md (context: the OpenAI CH paper, paper, whose Theorem 1.1 contradicts Theorem 4.1 under CH and whose Lean version passed the official comparator).

Three new reports: - astra-blind.md : a referee who saw ONLY Espíndola's two papers (not told about the CH paper). - astra-informed.md : walks the proof through the concrete CH class K. - opus-informed.md : independent Claude Opus investigator, same task as the previous one.

Read them, check the load-bearing claims against the sources yourself (especially: what thresholds.tex:359 asserts about the dense topology and its generators; whether the categorical model is a point of the double-negation subtopos; the "kappa^+-closed models are precisely the kappa^+-points" claim and its source in 1906.09169 (amalgamation hypothesis?); the "base rigidity" fact both informed reports use; the LS(K) vs accessibility-index ambiguity; the "modified morphisms" reply). Be adversarial: where could all three be wrong, e.g. by misreading his notation or topos-theoretic conventions (sites C vs C^op, which direction of maps covers go, what "point" means)? Note where the blind referee independently agrees or disagrees with the informed reports — independent agreement on the same step is the key signal.

Write an ADDENDUM (Markdown, under 1500 words) to the first-round report, for non-expert maintainers who may forward it to experts (set theory/AEC model theory, topos theory): 1. Summary (at most 7 sentences, neutral; no "Espíndola is wrong" language; say what was found and how confident). 2. The step(s) flagged, with exact quotes and file:line/PDF page, what is claimed, why it is doubted, and which reports flagged it (blind vs informed). 3. Other doubtful steps (ranked) and where the reports disagree. 4. Possible ways out (LS vs accessibility index; modified morphisms) and their status. 5. Questions for experts (short, concrete). 6. Method and limitations (AI-only; nothing checked by a person; topos-theoretic reasoning is where AI reading is least reliable). Write it as your final message.

Appendix D. Round-1 reports

Astra on objection E1

(1) **Short answer.** In the supplied manuscript, I find **no mismatch between Lemma 5.2 and Proposition 5.7 of the kind alleged in E1**. Two distinctions are decisive:

- Proposition 5.7 actually concludes $|I^M| < \Lambda$, where $\Lambda = \beth_{(2^{\aleph_1})^+}$. It does **not** conclude $|I^M| < \beth_{\delta^+}$.
- Lemma 5.2 prohibits a particular fixed sequence of compatible colors. The cardinal $|V_\delta|^+$ occurs in its proof as the size of an **auxiliary index set**, not as an upper bound on the index sort of every model.

The separate endpoint construction takes $I = V_{\omega_2}$ and produces the claimed model with uncountable B and total cardinality $\beth_{\omega_2}$. I reconstructed that construction and the intervening partition argument;

I found no gap relevant to E1. Thus, for these files, there is no identified typo or gap requiring repair, and E1 does not invalidate Theorem 1.1(ii).

Confidence: high for this scoped finding. There is a source-version uncertainty: the bound attributed to Proposition 5.7 in the thread differs from both the supplied PDF and TeX. I cannot determine whether Espíndola was reading another version.

I used only local files and `pdftotext`. No network access or file changes were made.

(2) **Detailed argument and source locations.**

First, the PDF numbering matters. The supplied PDF places the endpoint construction in **Section 4**, and the bound and categoricity argument in **Section 5**. Although its filename begins `04`, `04-endpoint.tex` is input before `03-categoricity.tex`; see paper.tex:13.

The PDF confirms:

Statement	Actual location
Theorem 1.1	p. 2
Lemma 4.2, cumulative-hierarchy cardinalities	p. 15
Theorem 4.5, two endpoint models	p. 18
Lemma 5.2, "No fixed pattern at all finite arities"	statement p. 19; proof pp. 20–21
Proposition 5.7, bound for uncountable base	p. 24
Theorem 5.10, eventual categoricity of this class	p. 26
Proposition 7.2	p. 30

Consequently, the filename-to-section shortcut in the task description does not apply to this supplied version.

The objection itself is recorded at x-thread.txt:14, with the further "interchange of successor and ב operator" observation at line 19.

The three different cardinal scales. The manuscript's standing notation is exactly

$$\delta = \omega_1, \qquad \eta = \delta^+ = \omega_2, \qquad S = 2^\omega/{=^*}, \qquad \xi = 2^\delta, \qquad \Lambda = \beth_{\xi^+}.$$

Its threshold function is exactly

$$h(\kappa) = \beth_{(2^\kappa)^+}, \qquad H(K) = h(\mathrm{LS}(K)).$$

Sources: 00-introduction.tex:193, lines 193–197, and line 21.

Espíndola correctly distinguishes $(\beth_\delta)^+$ from $\beth_{\delta^+}$. Using Lemma 4.2, which states

$$|V_{\omega+\gamma}| = \beth_\gamma \quad \text{for every ordinal } \gamma,$$

and $\omega + \delta = \delta$, we obtain $|V_\delta| = \beth_\delta$. Thus the following is a **derived comparison**, not a quotation:

$$|V_\delta|^+ = (\beth_\delta)^+ \le \beth_{\delta+1} < \beth_{\delta^+} = \beth_\eta < \beth_{\xi^+} = \Lambda.$$

The first strict inequality uses $\delta + 1 < \delta^+$; the last uses $\xi = 2^\delta \geq \delta^+ = \eta$, hence $\xi^+ > \eta$. No interchange of the two successor operations is legitimate or needed.

Under CH and $\mathrm{LS}(K) = \aleph_0$, the manuscript identifies

$$H(K) = \beth_{(2^{\aleph_0})^+} = \beth_{\omega_2} = \beth_\eta.$$

See Lemma 4.2, p. 15, and 05-conclusion.tex:3, lines 3–19.

Inventory of the requested bounds. The following records their occurrences and uses in the manuscript, including equivalent notation.

Quantity	Locations and role
$\beth_{\delta^+}$	This exact expression is not used in the manuscript's TeX. It is represented by $\beth_\eta$, with $\eta = \delta^+ = \omega_2$. The identification is in `00-introduction.tex:194` and `04-endpoint.tex:6`. Endpoint cardinality occurs in `04-endpoint.tex:62`, 335–358; the final identification with $H(K)$ is in `05-conclusion.tex:7–19`.
$(\beth_\delta)^+$	This exact expression is absent from the manuscript. It equals the auxiliary cardinal $\lvert V_\delta\rvert^+$, by Lemma 4.2. It is not stated as a bound on $\lvert I^M\rvert$.
$\lvert V_\delta\rvert^+$	Its sole explicit occurrence is `03-categoricity.tex:95`, in Lemma 5.2's choice of an auxiliary X. The associated inequalities are at lines 94 and 194. The collapse into V_δ is proved in Lemma 5.1, lines 28–74.
$H(K)$	Defined at `00-introduction.tex:21`; used in the proposed transfer at 49–50, the existence-bound discussion at 58–67, and Theorem 1.1 at 93 and 98–99. The proof identifies and compares it with $\beth_\eta$ and Λ at `05-conclusion.tex:7–19`. The consistency formulation subsequently writes the same threshold as $h(\aleph_0)$, notably at lines 24–41 and 69–71.
$\beth_{\omega_2}$	Explicitly occurs in `00-introduction.tex:5`, 7, 93 and 151, and `05-conclusion.tex:7`. Its constructive justification is the $\beth_\eta$ calculation in Lemma 4.2 and Theorem 4.5.

Quantity	Locations and role
$\Lambda = \beth_{(2^{\aleph_1})^+}$	This is the actual upper bound relevant to Proposition 5.7. Defined in `00-introduction.tex:94` and 193–197; recalled at `03-categoricity.tex:7–8`; used in the cardinal and partition estimates at 228–438, Proposition 5.7 at 440–484, and Theorem 5.10 at 602–626. It is compared with $H(K)$ in `05-conclusion.tex:12–19`, used in the consistency formulation at 24–41 and the joint-embedding/amalgamation argument at 106–132, and used for the tail application of Proposition 7.2 at `06-booleanization.tex:153–190`.

Source files: introduction, endpoint construction, categoricity argument, conclusion, Booleanization.

What Lemma 5.2 actually proves. Its statement is:

> There is no sequence $(s_j : j \geq 1)$ of elements of S such that
>
> $$T_k((s_{|J|})_{\emptyset \neq J \subseteq [k]}) \quad \text{holds for every } k \geq 1.$$

See Lemma 5.2, p. 19, and 03-categoricity.tex:76.

Here there is **one fixed color for each arity**. A passing test says that each decoded diagram is valid and that its restrictions agree exactly with the diagrams decoded at smaller arities.

The proof can be reconstructed as follows.

1. Suppose such a sequence exists. Decode diagrams $\mathcal{D}_k$. The tests give exactly
$$\mathcal{D}_k \restriction J = \mathcal{D}_{|J|} \qquad (\emptyset \neq J \subseteq [k]).$$
2. Choose a new linearly ordered set X with $|X| > |V_\delta|$; the manuscript suggests cardinality $|V_\delta|^+$.
3. For every finite nonempty $A \subseteq X$, copy $\mathcal{D}_{|A|}$ onto labels indexed by subsets of A. Exact restriction compatibility gives injective transition maps between their quotients, preserving tags and preserving **and reflecting** the relation.
4. Form their directed limit Q. Distinct elements of Q lie together in a finite stage. That stage contains a predecessor distinguishing them, and relation reflection ensures that the witness remains distinguishing in the limit. Thus the limit relation is extensional.
5. The distinguished singleton labels give an injection $X \to Q$. The tags give a function $Q \to \delta$ strictly increasing along the relation.
6. Lemma 5.1 collapses this tagged extensional relation injectively into V_δ, giving the contradiction
$$|X| \leq |Q| \leq |V_\delta|.$$

These steps are explicit at `03-categoricity.tex:85–196`, PDF pp. 20–21.

The crucial point is that X **is not required to be a subset of** I^M. Once the fixed diagrams exist, they can be copied over any set X. The role of $|V_\delta|^+$ is to make this independently constructed limit too large to collapse into V_δ.

Therefore Lemma 5.2 is not a theorem saying "every uncountable-base model has index size below $|V_\delta|^+$." The arithmetic observation in E1 is correct; that interpretation of the lemma is not supported by its statement or proof.

How Proposition 5.7 obtains the forbidden sequence. Its complete statement is:

> If $M \in K$ and B^M is uncountable, then $|I^M| < \Lambda$.

See Proposition 5.7, p. 24, and 03-categoricity.tex:440.

The bridge from a large model to Lemma 5.2 is substantive. It consists of Lemmas 5.4–5.6.

First fix row representatives. Every tuple t has a whole color profile $c_t : B^M \to S$. The manuscript calculates exactly

$$|S^{B^M}| \leq \delta^\delta = 2^\delta = \xi.$$

This uses CH, $|S| = \delta$, and $|B^M| \leq \delta$. Thus the partition argument colors tuples by profiles from a palette of size at most ξ, rather than merely by individual colors from S.

The cardinal estimates then establish:

$$\mathrm{cf}(\Lambda) = \xi^+ > \xi,$$

and that Λ is a strong limit, with both ξ^ζ and $(\xi^\zeta)^+$ below Λ whenever $\zeta < \Lambda$ is infinite. These facts follow from continuity and strict increase of the beth sequence at the regular limit ordinal ξ^+. See Lemma 5.4, p. 21; source lines 228–264.

I also checked the finite partition estimate, rather than assuming it:

- For arity one, $\mu = (\max(\xi, \chi))^+$ suffices to obtain a homogeneous set of size χ.
- For the induction step, choose a source cardinal ζ for the preceding arity and put exactly

$$q = \xi^\zeta, \qquad \mu = q^+ < \Lambda.$$

- The residual-set tree in the proof has at most q possible nodes. Hence some $x < q^+$ is never selected as a node's representative.
- Following the residual sets containing x produces ζ increasing representatives. Their profiles permit replacement of the last point of an $(m+1)$-tuple by x, reducing the coloring to arity m.
- At limit stages the residual contains x, so the branch does not disappear.

This supplies the claimed homogeneous subsets for each finite arity. See Lemma 5.5, pp. 22–23; source lines 267–390.

Lemma 5.6 then chooses profiles p_j recursively, retaining samples of unbounded cardinalities below Λ homogeneous for the first j selected profiles. The inequality $\xi < \mathrm{cf}(\Lambda)$ ensures that some next profile occurs for unboundedly many target sizes. Consequently, for every k, there is a k-tuple t_k whose j-subtuples all have profile p_j. These tuples need not be nested. See pp. 23–24; source lines 398–437.

Finally, each t_k has a countable exceptional set E_k. Their countable union is countable. Since B^M is uncountable, choose b outside that union and put $s_j = p_j(b)$. Every finite test required by Lemma 5.2 now holds. This contradiction proves Proposition 5.7.

Thus the actual implication is:

$$\begin{aligned} |I^M| \geq \Lambda, \quad B^M \text{ uncountable} \\ \implies \text{fixed profiles with finite witnesses} \\ \implies \text{one coordinate giving a forbidden color sequence.} \end{aligned}$$

There is no substitution of $\beth_{\delta^+}$ for $(\beth_\delta)^+$.

A useful additional check is that the same selection argument cannot simply be run at the endpoint. By the same beth-cofinality calculation,

$$\operatorname{cf}(\beth_\eta) = \eta \leq \xi.$$

So the crucial strict inequality "palette size < cofinality" is unavailable there. This explains mathematically why the proof uses the larger Λ.

The endpoint construction independently gives size $\beth_{\omega_2}$. This is Theorem 4.5, p. 18, proved using Lemmas 4.1–4.4. It does not derive existence from Proposition 5.7.

The construction's essential steps are as follows.

1. **Represent η many ranks modulo countable exceptions.** For each $\alpha < \eta = \delta^+$, choose an injection $j_\alpha : \alpha \to \delta$, and define exactly

   $$f_\alpha(\beta) = \sup\{f_\gamma(\beta) + 1 : \gamma < \alpha,\ j_\alpha(\gamma) < \beta\}, \qquad \sup \emptyset = 0.$$

   For each $\beta < \omega_1$, only countably many terms occur, so regularity of ω_1 keeps the value below δ. For fixed $\gamma < \alpha$, the inequality $f_\gamma(\beta) < f_\alpha(\beta)$ holds whenever $\beta > j_\alpha(\gamma)$. Thus its failure set $E_{\gamma,\alpha}$ is countable. This is Lemma 4.1, p. 15; 04-endpoint.tex:17, lines 17–53.

2. **Take the actual index set $D = V_\eta$.** For every finite nonempty $X \subseteq D$, construct a countable $D_X \subseteq D$ containing X, containing all D_Y for nonempty $Y \subseteq X$, and having extensional actual membership. Repeatedly adding one witness from each symmetric difference makes membership extensional while preserving countability. Transitivity of V_η keeps every witness in D. See Lemma 4.3, p. 16; source lines 98–161.

3. **Use the approximate ranks as local tags.** Enumerate each D_X by $d_X : \omega \to D_X$, with $d_{\{x\}}(0) = x$. For a tuple $t = (x_1, \dots, x_k)$, put

   $$q_t(J, i) = d_{X_J}(i).$$

   Its diagram uses equality and membership of these values; its tag at coordinate β is $f_{\operatorname{rank}(q_t(\ell))}(\beta)$.

   The exceptional set is exactly

   $$F_X = \bigcup_{\substack{x,y \in D_X \\ y \in x}} E_{\operatorname{rank}(y),\operatorname{rank}(x)}.$$

   It is countable because D_X is countable. Moreover, $F_Y \subseteq F_X$ whenever $\emptyset \neq Y \subseteq X$.

4. **Verify all parts of the full test.** Outside F_X, every membership edge has strictly increasing tags. Each quotient is the extensional relation on the corresponding D_Y. Distinguished singleton labels denote the distinct original tuple entries. Restricting labels gives exactly the separately constructed subtuple diagram because enumerations were fixed for underlying finite sets.

Thus the full test, including validity of every restriction, passes. See Lemma 4.4, p. 17; source lines 164–278.

5. **Realize these colors as an object.** Take $B = \delta \times \mathbb{Q}$ in lexicographic order. It is a dense order without endpoints, has cardinality $\aleph_1$, and has countable bounded intervals. Enumerate its points as b_β, independently of their order. The coding maps encode the diagrams, including invalid diagrams at exceptional coordinates. Lemma 3.8 realizes representative binary sequences in the torsor structure. The manuscript obtains exactly

$$E_t(\mathbf{z}) \subseteq \{b_\beta : \beta \in F_X\}.$$

Hence every tuple has a countable exceptional set, as the object definition requires. See source lines 281–333, PDF pp. 17–18.

The resulting model satisfies exactly

$$|I^{M_{\mathrm{u}}}| = |V_\eta| = \beth_\eta, \qquad |B^{M_{\mathrm{u}}}| = \aleph_1.$$

For the second model, take an index set of the same cardinality, $B = \mathbb{Q}$, and zero bit sequences. The object condition is automatic when B is countable. Lemma 3.5 gives total size $\beth_\eta$ for both models, and an isomorphism cannot change the cardinality of the named base sort. See Theorem 4.5, p. 18; source lines 335–365.

Why this does not conflict with collapse into V_δ. The object condition permits a different countable exceptional set for every tuple. It does not supply one coordinate at which all tuples pass. This is explicit in 02-aec.tex:280, lines 280–300.

Indeed, in this endpoint construction no single coordinate β could make all membership diagrams valid. Applying validity to pairs of ordinals $\gamma < \alpha < \eta$ would require $f_\gamma(\beta) < f_\alpha(\beta)$ for all such pairs—an increasing sequence of length $\eta = \delta^+$ inside δ, which is impossible.

Local validity outside varying countable sets therefore does not yield the global tagged relation to which the collapse argument would apply. Proposition 5.7 obtains the much stronger fixed-pattern configuration only after its partition argument.

Espíndola's description of the model size is accurate. His thread says the class has "in particular two sorts," B and I. The manuscript calls them its "two principal sorts"; it also has group and torsor carriers. Thus the thread need not be read as claiming that there are only two sorts.

Lemma 3.4 proves $|B| \leq \aleph_1$: above any fixed point, either there is a countable cofinal set, or an increasing ω_1-sequence must be cofinal because a point above it would create an uncountable bounded interval. The reversed argument handles the other side.

Lemma 3.5 states exactly

$$\|M\| = \max(|I^M|, |B^M|, \aleph_0).$$

There are at most that many tuples and fibers; row fibers have size $|B|$, and column fibers are countable. Since every allowed B is infinite, the $\aleph_0$ term is redundant. Thus Espíndola's abbreviated formula $\max(|I^M|, |B^M|)$ gives the same cardinality.

See Lemmas 3.4–3.5, p. 9; 02-aec.tex:167, lines 167–209.

Consequences for the theorem, as distinct from consequences for a proof. The endpoint construction supports part (ii) directly. Proposition 5.7 supports the higher categoricity tail: at total size $\mu \geq \Lambda$, the size formula forces $|I^M| = \mu$, so the base must be countable. Lemma 5.9 identifies all countable-base models with equally sized index sorts, yielding Theorem 5.10. See PDF pp. 24–26.

Accordingly:

- The supplied text exhibits no E1 gap to classify as a typo or a missing argument.
- A hypothetical error in Proposition 5.7 would threaten the proof of the tail assertion, part (iii); it would not by itself disprove the independently constructed endpoint models in part (ii).
- If another version really asserted $|I^M| < \beth_{\delta^+}$ for every uncountable-base model of this same class, that assertion would conflict with the endpoint construction above. Deciding which argument failed would require that version.

(3) **Confidence in the conclusions.**

Conclusion	Confidence and reason
The supplied Proposition 5.7 uses $\Lambda = \beth_{(2^{\aleph_1})^+}$.	**High.** Direct agreement between PDF p. 24, standing definitions, and TeX.
$\|V_\delta\|^+ = (\beth_\delta)^+ < \beth_{\delta^+}$.	**High.** Follows from the hierarchy calculation and ordinary cardinal arithmetic.
Lemma 5.2's auxiliary cardinal is not a universal model-size bound.	**High.** Its statement and the construction of a fresh X explicitly establish the different role.
The passage from Lemma 5.2 to Proposition 5.7 has no identified E1 gap.	**High.** The profile palette, finite partition induction, cofinality selection, and countable-exception argument fit together without exchanging the bounds.
The endpoint construction supplies the two models claimed in Theorem 1.1(ii), under the stated definitions and CH.	**High.** The rank approximation, countable closures, restriction compatibility, array realization, cardinality, and nonisomorphism checks can be reconstructed directly.
Espíndola's base-size and total-size description is accurate.	**High.** It agrees with Lemmas 3.4–3.5, with a redundant $\aleph_0$ term omitted.
Why the thread attributes a different bound to Proposition 5.7.	**Low / undetermined.** No historical version establishing that attribution was supplied.

(4) **What I could not determine.** I cannot establish which manuscript version Espíndola read, whether an earlier version contained the alleged bound, or whether revisions preceded the supplied files. The printed date alone does not resolve that history.

This investigation reconstructs the arguments relevant to E1; it is not a complete independent certification of every AEC axiom or of the entire manuscript. I have not assessed E2 or Espíndola's categorical-equivalence argument here.

I have not used Lean compilation or the pending comparator result as evidence. Nothing in this report determines whether the formal statement perfectly matches the manuscript; the E1 findings above stand on the manuscript's definitions and proofs.

(5) **Concrete questions for a domain expert.**

1. **Version identification:** Can the objector provide the exact PDF or commit in which Proposition 5.7 concludes $|I^M| < \beth_{\delta^+}$? The supplied statement on p. 24 concludes $|I^M| < \Lambda$.
2. **Lemma 5.2's scope:** Is there any reason its auxiliary X must lie inside the original model's index sort? The copying construction on pp. 20–21 appears to impose no such requirement.
3. **The partition bridge:** Is there a specific failure in the residual-tree argument of Lemma 5.5 or the unbounded-sample selection of Lemma 5.6, pp. 22–24? These are the exact steps justifying the larger Λ.
4. **The endpoint object:** In Lemmas 4.1–4.4, pp. 15–17, does any required validity or restriction condition fail outside the displayed countable F_X? Identifying such a failure would directly challenge the uncountable-base model.
5. **The alleged collapse contradiction:** Can E1 produce a common valid coordinate or a fixed color sequence for the endpoint model without assuming the partition hypothesis $|I^M| \geq \Lambda$? That is the additional information needed to apply Lemma 5.2.

Astra on the Lean formalization

(1) **Short answer.** The supplied Lean theorem asserts the same mathematical conclusion as the manuscript's Theorem 1.1: under CH, there is a countable-language AEC with Löwenheim–Skolem number $\aleph_0$, two nonisomorphic models of size $\beth_{\omega_2} = H(K)$, and exactly one isomorphism type at every cardinal at least $\Lambda = \beth_{(2^{\aleph_1})^+}$. I found no weakened AEC axiom, vacuous definition of categoricity, or substitution of the wrong endpoint cardinal.

The Lean proof uses **three different cardinals for three different purposes**:

Purpose	Cardinal
Auxiliary contradiction in the counterpart of Lemma 5.2	$\lvert V_{\omega_1}\rvert^+ = (\beth_{\omega_1})^+$
Size of the two explicitly constructed endpoint models	$\beth_{\omega_2}$
Upper bound for index sorts of uncountable-base models, in the counterpart of Proposition 5.7	$\Lambda = \beth_{(2^{\aleph_1})^+}$

There is also a checkable discrepancy between E1's description and the supplied manuscript: **Proposition 5.7 actually concludes** $|I^M| < \Lambda$, not $|I^M| < \beth_{\delta^+}$.

Assuming acceptance of the solution proof with the permitted standard axioms, and agreement on the mathematical interpretation of its definitions, the result bears directly on E1: it establishes the disputed endpoint models at $\beth_{\omega_2}$. This does not certify every sentence of the manuscript's prose proof or settle E2.

I used only local files, made no edits, and did not run Lean or receive the separate comparator result.

(2) **Detailed argument and exact locations.**

A source-location correction is necessary. In this checkout, paper/build/paper.tex:13–14 inputs `04-endpoint` **before** `03-categoricity`. Running `pdftotext` on the supplied PDF confirms:

PDF location	TeX source
Theorem 1.1, p. 2	`00-introduction.tex:87–100`
Section 4; endpoint Theorem 4.5, p. 18	`04-endpoint.tex:335–365`
Lemma 5.2, pp. 19–21	`03-categoricity.tex:76–197`
Proposition 5.7, p. 24	`03-categoricity.tex:440–485`
Proposition 7.2, p. 30	`06-booleanization.tex`

Thus the filename-to-section mapping in the question does not match the supplied build order. All numbered manuscript citations below follow the PDF.

The exact theorem being proved. The solution's declaration is:

```
theorem main (hCH : CH.{u}) :
    ∃ (L : ℕ → Type), Countable (Σ n, L n) ∧
      ∃ K : ClassData.{u} L, K.IsAEC ∧ K.HasLSNumber ℵ₀ ∧
        endpoint = hanf (ℵ₀ : Cardinal.{u}) ∧ K.TwoModels endpoint ∧
        ∀ μ, tailThreshold ≤ μ → K.Categorical μ := by
```

Its proof supplies the constructed language, class, AEC verification, LS verification, cardinal equality, endpoint models, and categoricity tail separately. See Main.lean:233–240.

The challenge file ends in `sorry`, as stipulated. The JSON identifies `OAI.ModelTheory.Categoricity.Main` as the solution module and permits only `propext`, `Quot.sound`, and `Classical.choice`. I compared the source text: the entire definitions block in the challenge is identical to the corresponding block in `Model.lean`, and the two `main` theorem types are identical. This is a source comparison, not a substitute for the official comparator. See CHObstruction.lean:13–101, CHObstruction.json:1–13.

In mathematical language, its definitions mean the following.

- **Language and structures.** $L(n)$ is the set of n-ary relation symbols. The disjoint union $\coprod_{n<\omega} L(n)$ is countable. A structure consists of a set A, represented by `ZFSet.{u}`, and an interpretation $r^M \subseteq A^n$ for each $r \in L(n)$. Every arity is finite.
- **Substructure.** M is a substructure of N when its carrier is literally a subset of N's carrier and every relation on old tuples agrees in both directions.
- **Isomorphism.** An isomorphism is a bijection of carriers preserving and reflecting every relation. It is neither mere equicardinality nor a separately chosen restrictive class of isomorphisms.
- **Class data.** K consists of a predicate selecting structures and a binary relation $M \leq_K N$, represented by `objects` and `strong`.

These definitions appear in Model.lean:26–52.

`K.IsAEC` requires all the following:

1. Strong substructure relates only objects of K.
2. It is reflexive on objects, antisymmetric, and transitive.
3. Strong substructure implies induced relational substructure.
4. Objects are closed under arbitrary language isomorphisms.
5. Strong substructure is invariant under isomorphisms: compatible isomorphisms of the smaller and larger structures transport a strong inclusion to a strong inclusion.

6. **Coherence:** if $M_0 \subseteq_L M_1$, $M_0 \leq_K M_2$, and $M_1 \leq_K M_2$, then $M_0 \leq_K M_1$.
7. **Unions and smoothness:** for every nonzero ordinal γ and increasing chain $(M_i)_{i<\gamma}$, there is an object U whose carrier is exactly the union, whose relations restrict correctly to each stage, and with $M_i \leq_K U$. If all stages are strong substructures of another object N, then $U \leq_K N$.
8. A downward Löwenheim–Skolem bound exists.

There is no cofinality restriction on these chains. The definition of `IsUnion` requires both inclusion of every stage and coverage of every element of the union. Finitary relations and the chain condition make this the usual relational union. See Model.lean:44–46, Model.lean:66–88.

Although the public interface defines strong **inclusions**, its associated strong embeddings have the usual meaning: an injective language embedding $f : M \to N$ whose induced image satisfies $f[M] \leq_K N$. The isomorphism-invariance axiom permits transport between inclusions and this formulation.

The LS definitions say:

$$\theta \geq \aleph_0, \qquad \theta \geq |L|,$$

and, for every $M \in K$ and every subset $A \subseteq |M|$, there is $N \leq_K M$ containing A, with

$$\|N\| \leq |A| + \theta.$$

`HasLSNumber θ` additionally requires that θ be no greater than **every** LS bound. Thus `HasLSNumber ℵ₀` asserts the least LS number, not merely a convenient upper bound. Since θ is infinite, the separate requirements $\theta \geq \aleph_0$ and $\theta \geq |L|$ agree with the manuscript's requirement $\theta \geq |L| + \aleph_0$. See Model.lean:58–64.

`Categorical μ` means:

- at least one object has carrier cardinality μ; and
- any two such objects are language-isomorphic.

`TwoModels μ` means that two objects of cardinality exactly μ exist and admit no language isomorphism. See Model.lean:90–97.

The cardinal definitions are exactly:

```
def CH : Prop := (2 : Cardinal.{u}) ^ ℵ₀ = ℵ_ 1

def hanf (κ : Cardinal.{u}) : Cardinal.{u} :=
  beth (succ ((2 : Cardinal.{u}) ^ κ)).ord

def endpoint : Cardinal.{u} := beth (ω_ 2)

def tailThreshold : Cardinal.{u} :=
  beth (succ ((2 : Cardinal.{u}) ^ (ℵ_ 1))).ord
```

Here `succ` acts on a **cardinal**, and `.ord` takes its initial ordinal before it is used as the beth index. Thus:

- CH is $2^{\aleph_0} = \aleph_1$.
- `hanf κ` is $h(\kappa) = \beth_{(2^\kappa)^+}$.

- `endpoint` is $ב_{\omega_2}$.
- `tailThreshold` is $\Lambda = ב_{(2^{\aleph_1})^+}$.

In particular, `ω_ 2` denotes ω_2, not ω^2 or the ordinal 2. There is no separate Lean definition named $H(K)$: because the theorem establishes $\mathrm{LS}(K) = \aleph_0$, its `hanf ℵ₀` is exactly the manuscript's $H(K) = h(\mathrm{LS}(K))$. The manuscript writes:

$$h(\kappa) = ב_{(2^{\kappa})^+}, \qquad H(K) = h(\mathrm{LS}(K)).$$

See Model.lean:101–107, 00-introduction.tex:20–26.

The theorem is polymorphic in the universe level u; carriers, cardinals, and chain indices use the corresponding universe. Each carrier is an arbitrary `ZFSet.{u}`, rather than a subset of a single prescribed carrier. The tail quantifier ranges over every `Cardinal.{u}`. This is the usual universe-relative formulation of set-sized mathematics.

Comparison with Theorem 1.1 and checks for altered meaning. The three numbered conclusions of Theorem 1.1, p. 2, match the Lean conjunction directly. The manuscript's final sentence concerning $\lambda = \Lambda^+$ is established separately by `prescribed_transfer_fails`; `endpoint_lt_tailThreshold` proves the strict separation of the endpoint and the tail. See 00-introduction.tex:87–100, Main.lean:242–264.

The concrete class also follows the manuscript's construction:

- `Arrays.Data` contains nonempty index and base sets, a dense linear order without endpoints on the base, countable bounded intervals, and a bit array for every nonempty finite injective tuple.
- Its carrier contains tagged index, base, finite-support group, fixed countable group, row, and column points. The relational language encodes the sorts, projections, order, incidence, bits, group operations, and torsor actions.
- `Data.IsObject` requires that, for **each tuple**, only countably many base coordinates fail its full subsequence test.
- `Map.Strong` requires convexity of the base image and successful tests at **every new base coordinate**, using the images of the old row representatives.

See Arrays.lean:61–144, ArrayEmbeddings.lean:40–94. Compare the manuscript's basic structures and strong inclusions at 02-aec.tex:59–165 and its object definition at 02-aec.tex:269–300.

Several details specifically address possible misformalization:

Potential issue	Finding
Empty class or absent tail models	Existence is part of `Categorical`; `Existence.models` constructs a model at every infinite cardinal.
Uniqueness obtained by weakening isomorphism	`RelModel.Iso` preserves and reflects every relation.
Extra named torsor origins changing isomorphism types	Origins are coordinates in presentations, not additional relation symbols. Row and column translations are allowed in maps.

Potential issue	Finding
Only canonical carrier codes counted as objects	`classData.objects` includes every relational structure admitting an isomorphic array presentation.
Omitted smoothness or short-chain restriction	Both are explicitly covered; the implementation proves unions for arbitrary nonempty directed families.
LS number merely asserted	A small strong submodel is constructed for every subset, then minimality at $\aleph_0$ follows from the definition.
Wrong meaning of model size	Size is the whole carrier; Lean proves `#M.Carrier = max (max #M.I #M.B) ℵ₀`.

Relevant proofs are Main.lean:49–83, AEC.lean:64–162, AEC.lean:291–424, and Subarrays.lean:134–167.

The tests themselves are substantive. Diagrams contain an equivalence relation, a membership-like relation compatible with that equivalence, tags below ω_1 decreasing along membership, extensionality, and distinct distinguished roots. `Passes` requires validity and the correct code for **every** nonempty increasing subsequence, including the full tuple. See Diagrams.lean:60–75, Colors.lean:146–153.

For countable bases, the object condition is automatically satisfied because every exceptional subset is countable. This is an explicit feature of both constructions. Tail uniqueness then requires an actual isomorphism argument: finite row and column translations absorb the difference between bit arrays over a countable base. See Arrays.lean:24–52, ArrayEmbeddings.lean:220–237.

I read all 15 proof modules. A source scan found no `sorry`, `admit`, or additional axiom declarations in them; their import chain ultimately reaches `Mathlib`. This observation does not establish the compiled theorem's axiom dependencies—that remains a kernel/comparator check.

The counterpart of Lemma 5.2. The manuscript's statement is a prohibition on a fixed sequence of colors:

$$T_k((s_{|J|})_{\emptyset \neq J \subseteq [k]}) \quad \text{holds for every } k \geq 1.$$

Lemma 5.2 says that no such sequence exists. Its conclusion is not a model-size bound. The smaller cardinal occurs inside its contradiction proof, which chooses an auxiliary ordered set X with $|X| > |V_\delta|$, for example of size $|V_\delta|^+$. See Lemma 5.2, pp. 19–21; 03-categoricity.tex:76–106.

The corresponding Lean chain is:

Manuscript step	Lean location
Collapse an extensional tagged relation into V_δ	`TaggedCollapse.embedding`, Collapse.lean:64–74
Preserve extensionality and tags in the diagram limit	`TaggedLimit.embedding`, Collapse.lean:294–324
Embed an arbitrary ordered root set into V_{ω_1}, assuming coherent diagrams at all arities	`NoPattern.rootsEmbedding`, Diagrams.lean:272–279

Manuscript step	Lean location
Choose a root set too large for that injection	`NoPattern.impossible`, Diagrams.lean:285–293
Obtain coherent diagrams from fixed colors passing all tests	`Tests.no_pattern`, Colors.lean:173–201

The decisive code is exactly:

```
let V := ZFSet.vonNeumann (ω_ (1 : Ordinal.{u}))
let J := (succ V.card).ord.ToType
```

It obtains an injection from J into a cardinality-preserving small copy of V, derives

```
have hle : succ V.card ≤ V.card := by
```

and contradicts the strict successor inequality.

Thus E1 correctly identifies the smaller cardinal used **inside this argument**. Its role is to disprove the existence of a fixed coherent pattern. The root set J is newly chosen for that contradiction; it is not asserted to be the original model's index sort.

The counterpart of Proposition 5.7. The supplied proposition reads:

> If $M \in K$ and B^M is uncountable, then $|I^M| < \Lambda$.

Its section recalls, exactly,

$$\delta = \omega_1, \qquad \xi = 2^{\delta}, \qquad \Lambda = \beth_{\xi^+}.$$

See Proposition 5.7, p. 24; 03-categoricity.tex:4–16, 03-categoricity.tex:440–456.

The direct Lean counterpart is:

```
theorem Data.uncountable_index_bound (hCH : CH.{u}) (hM : M.IsObject hCH)
    (hB : ¬ Countable M.B) : #M.I < tailThreshold.{u} :=
  TailBound.index_lt hCH M.base_bound hB M.colors hM
```

See Arrays.lean:152–154.

The underlying `TailBound.index_lt` argument is:

1. The set of whole color profiles $B \to S$ has cardinality at most $\xi = 2^{\aleph_1}$.
2. Assuming $|I| \geq \Lambda$, finite partition estimates select a fixed profile p_j for each positive arity j.
3. For each k, a finite witness tuple has those profiles on its subsequences. These witness tuples need not be nested.
4. Only countably many witness tuples are selected. Their exceptional sets therefore have countable union.
5. Uncountability of B supplies one coordinate b outside that union.
6. Evaluating p_j at b gives the fixed sequence forbidden by `Tests.no_pattern`.

See TailBound.lean:83–122.

The partition argument uses that Λ is a strong limit and

$$\mathrm{cf}(\Lambda) = \xi^+ > \xi.$$

These facts are proved in Partition.lean:267–288; the selection of finite patterns is proved in Partition.lean:348–442.

Consequently, the passage from Lemma 5.2 to Proposition 5.7 does not require an equality between the smaller contradiction cardinal and Λ. It uses a separate partition argument to produce an absolutely forbidden pattern. The manuscript explicitly explains these quantifiers in Remarks 5.3 and 5.8, pp. 21 and 24.

Where the endpoint models actually come from. Their existence is established by a separate construction:

```
abbrev V := ZFSet.vonNeumann (ω_ (2 : Ordinal.{u}))
abbrev D := Shrink.{u} V.{u}
```

`Shrink` here is equipped with an equivalence to V; it changes the type presentation while preserving cardinality. The theorem

```
theorem cardinal : #D.{u} = endpoint.{u} := by
```

uses `ZFSet.card_vonNeumann` and the comparison between `preBeth` and `beth` to establish $|D| = \beth_{\omega_2}$. See EndpointDiagrams.lean:17–20, EndpointDiagrams.lean:168–175.

The construction uses an ω_2-indexed family of functions $\omega_1 \to \omega_1$, increasing modulo countable exceptional sets. The manuscript writes:

$$f_\alpha(\beta) = \sup\{f_\gamma(\beta) + 1 : \gamma < \alpha,\ j_\alpha(\gamma) < \beta\}, \qquad \sup \emptyset = 0.$$

Lean implements this recursion in `Domination.values` and proves `countable_exceptions`. See Lemma 4.1, p. 15; 04-endpoint.tex:17–54, Collapse.lean:80–149.

For each finite tuple from V_{ω_2}, countable membership diagrams receive local tags below ω_1. Their tests fail at only countably many coordinates **for that tuple**. This is proved by `EndpointDiagrams.countable_failures`. It does not assert a single coordinate at which every tuple passes. See EndpointDiagrams.lean:93–166.

Finally:

- the uncountable base is $\omega_1 \times_{\mathrm{lex}} \mathbb{Q}$, of cardinality $\aleph_1$;
- the index set is D, of cardinality $\beth_{\omega_2}$;
- the whole carrier has cardinality $\beth_{\omega_2}$;
- another model of that same total cardinality has base $\mathbb{Q}$;
- an arbitrary relational isomorphism would preserve the base predicate, contradicting the different base cardinalities.

These are the explicit witnesses in Main.lean:132–227. The manuscript gives the corresponding construction in Theorem 4.5, p. 18.

The quantifier distinction matters: tuple-dependent countable exceptions permit these endpoint models, while a fixed pattern valid at all finite arities would permit the auxiliary collapse contradiction. The proof does not interchange those assertions.

Conditional consequence for E1. Espíndola's supplied words identify "one interchange of successor and ב operator" and suggest that the Lean statement formalizes something else. See x-thread.txt:14–19.

The inspected code keeps those operations distinct. Under the stipulated successful kernel check, it proves the existence of the two endpoint models at $ב_{\omega_2}$, together with the AEC and tail properties. Given the semantic correspondence described above, E1's proposed restriction of the construction to $(ב_{\omega_1})^+$ would therefore not invalidate this formalized theorem.

This conclusion has a specific scope:

- A gap in a written proof can coexist with an independently correct formal proof.
- Identifying a gap would not by itself show that the theorem is false.
- Kernel acceptance establishes the encoded proposition; interpreting that proposition as the intended mathematics still requires definition review.
- The formalization contains no proof of Proposition 7.2 or the separate eventual-categoricity theorem. Its scope note explicitly excludes both. See 240.md:7–11.

(3) **Confidence levels.**

Conclusion	Confidence	Reason
The challenge and solution have identical public definitions and `main` types	**High**	Direct source-text comparison.
The theorem statement faithfully expresses Theorem 1.1	**High**	Its quantifiers, AEC conditions, LS minimality, existence clauses, isomorphisms, and cardinals match explicitly.
No vacuity or weakened public definition was found	**High**	Explicit existence witnesses and complete listed AEC axioms address the principal failure modes.
The concrete implementation corresponds to the manuscript's construction	**Medium**	The structures, tests, maps, and major proof steps align, but this remains a manual mathematical correspondence assessment.
The three cardinal roles listed above are distinct in the Lean proof	**High**	They occur explicitly in the contradiction, endpoint construction, and tail-bound declarations.
Proposition 5.7 in the supplied PDF concludes $\|I^M\| < \Lambda$	**High**	Confirmed directly in PDF text and TeX.
An accepted proof with these meanings bears directly on E1	**High, conditional**	Its conclusion includes precisely the disputed endpoint existence claim.

(4) **What I could not determine.**

- Whether the separate comparator has accepted the solution and confirmed its axiom dependencies. I make no claim about its result.
- Whether the supplied manuscript is exactly the version Espíndola was discussing. The discrepancy in Proposition 5.7's stated bound makes this a material provenance question.
- Whether every step of the manuscript's prose proof is complete. This investigation traced the formal theorem and relevant proof mechanisms; it does not certify the entire prose argument.
- A formal metatheoretic translation of this Lean development into ZFC, including the manuscript's relative-consistency conclusion.
- Whether Proposition 7.2 applies to the modified category invoked in E2, or whether Espíndola's Theorem 4.1 has a repair. Those questions are outside this formalization.

(5) **Concrete questions for a domain expert.**

1. **Definition match:** Does `Model.lean:58–97` omit any requirement of the AEC convention intended in this disagreement? In particular, is there any substantive issue with its formulation of isomorphism invariance, coherence, or chain smoothness?
2. **Core of E1:** In Lemma 5.2 and `Diagrams.lean:272–293`, is there any reason the auxiliary ordered set cannot be chosen independently after obtaining the fixed coherent diagrams?
3. **Endpoint construction:** Do `Domination.countable_exceptions` and `EndpointDiagrams.countable_failures` accurately capture the manuscript's use of actual ranks below ω_2 with local tags below ω_1?
4. **Partition bridge:** Does Proposition 5.7's use of separate finite witnesses match `FinitePatterns.exists_patterns` and `TailBound.index_lt`, without requiring a common infinite homogeneous subset?
5. **Version identification:** Can the parties identify the exact text in which Proposition 5.7 allegedly states $|I^M| < \beth_{\delta^+}$? The supplied PDF instead states $|I^M| < \Lambda$, on p. 24.

Astra on objection E2

(1) Short answer. Proposition 7.2 targets an assertion that appears explicitly in Espíndola's written proof: that the large categorical model supplies a point of the double-negation subtopos. It gives a concrete obstruction **for the manuscript's category with its original strong embeddings and its canonical model point**. It does not, by itself, establish the same obstruction for a category with genuinely different morphisms. Espíndola's reply therefore raises a relevant distinction, but the supplied sources do not identify sufficiently precisely how his proposed modification applies at this step.

The theorem statements nevertheless conflict. Espíndola's Theorem 4.1 covers the AEC claimed in the manuscript's Theorem 1.1, with threshold

$$\beth_{(2^{\aleph_0})^+} = \beth_{\omega_2} \qquad \text{under CH.}$$

The manuscript claims noncategoricity exactly there and categoricity above the larger threshold

$$\Lambda = \beth_{(2^{\aleph_1})^+}.$$

Thus **Theorem 1.1 and Theorem 4.1 cannot both be true in ZFC+CH**. This identifies incompatible mathematical claims; it does not determine which proof fails.

I used only the supplied local materials, checked numbering with `pdftotext`, and changed no files.

(2) Detailed argument. The two issues—identifying a faulty proof step and comparing theorem statements—require separate arguments.

What Espíndola's proof actually says.

Theorem 4.1 reads:

> "Let $\mathcal{K}$ be an AEC. If $\mathcal{K}$ is categorical in some $\lambda \geq \beth_{(2^{LS(\mathcal{K})})^+}$, then $\mathcal{K}$ is λ'-categorical for every $\lambda' \geq \beth_{(2^{LS(\mathcal{K})})^+}$."

This is arXiv:2301.13167v1, thresholds.tex:355, Theorem 4.1, PDF p. 6.

Its proof opens by taking the double-negation subtopos of $\mathbf{Set}^{\mathcal{K}_\kappa}$, with $\kappa = LS(\mathcal{K})$. It describes generators for the dense topology, asserts that the resulting sheaf topos is Boolean and is the double-negation subtopos, and then says:

> "Clearly the model of size λ must be one of its points".

That assertion occurs **before** the subsequent passage to a slice topos and a newly defined κ^+-AEC. The point is used to select a type and then an isolating subobject $S \rightarrowtail 1$. It is therefore an operative step in the proof. (arXiv:2301.13167v1, thresholds.tex:359, PDF p. 7.)

What Proposition 7.2 establishes.

Section 7 explicitly fixes a small skeleton $\mathcal{C}_0$ of countable models **with strong embeddings** and the presheaf topos

$$\mathcal{D} = \mathrm{Set}^{\mathcal{C}_0}.$$

Its canonical point satisfies, exactly as written,

$$p_H^*(X) = \operatorname*{colim}_{(C,c)\in \mathcal{C}_0/H} X(C), \qquad p_H^*(y_A) = \mathrm{Hom}_K(A,H).$$

The section also argues that internal size and underlying cardinality agree for this category, including at $\aleph_0$. (06-booleanization.tex:92–149, PDF pp. 29–30.)

For a model H with countable base, choose a countable strong map $a : A \to H$ whose image contains all of B^H. The proof defines

$$R_A = \{r : A \to D \text{ in } \mathcal{C}_0 : r[B^A] \subsetneq B^D\} \subseteq y_A.$$

Lemma 7.1 supplies a countable strong extension adding a base point after every arrow out of A. Consequently R_A satisfies the stated density condition:

$$\text{for every } f : A \to D \text{ there is } g : D \to E \text{ with } gf \in R.$$

However, a cannot factor through an arrow $r \in R_A$: an additional base point in D would have to map injectively into $B^H = a[B^A]$, colliding with an existing image. Thus $p_H^*(R_A) \to p_H^*(y_A)$ is not surjective, although sheafification inverts the dense inclusion. This obstructs factorization of p_H through the double-negation subtopos. (06-booleanization.tex:153–190, Proposition 7.2, PDF p. 30.)

This argument is precise about its scope:

- The nonfactorization argument for a countable-base H uses the countable-extension lemma and the downward Löwenheim–Skolem property.

- Its application to **every model on the categoricity tail** additionally uses Proposition 5.7 and Theorem 5.10.
- It does not show that the Booleanization has no points, or exclude another construction assigning a different point to a model. Section 7.3 expressly acknowledges those limitations. (06-booleanization.tex:193–207, PDF p. 30.)

How the "different morphisms" reply changes the comparison.

Espíndola's exact reply is:

> "I use an axiomatization of a category of models categorically equivalent to a modified AEC (same models, different morphisms) which do not affect the categoricity spectrum."

(x-thread.txt:15.)

Two different operations must be distinguished.

An equivalent presentation of the original category does not remove the obstruction. An equivalence transports representable functors, the corresponding presheaf topos, its double-negation subtopos, and the canonical point represented by the model's hom-functor. A change of notation or an equivalent axiomatization alone cannot turn this nonfactorization into factorization.

Changing morphisms can remove the particular obstruction. Even when objects, their cardinalities, and their isomorphism classes remain unchanged, the available arrows determine cosieves, density, and the hom-functor defining the canonical point. For example, restricting morphisms could make the extensions used in Lemma 7.1 unavailable. Proposition 7.2 would then require a new proof for that category.

The supplied papers contain relevant distinctions:

- The earlier paper describes an axiomatization equivalent to the **original AEC**, with unchanged categoricity spectrum. (arXiv:1906.09169v7, scc3.tex:288.)
- The thresholds paper's Section 3 describes a predicate expansion giving an **isomorphic AEC**. (arXiv:2301.13167v1, thresholds.tex:304.)
- It also explicitly introduces categories with altered morphisms: for example, $\mathcal{K}^{B}_{\geq\kappa,<\lambda}$, using κ-Boolean homomorphisms, and later $\mathcal{K}^{B,r}_{\geq\chi,<\lambda}$. (arXiv:2301.13167v1, thresholds.tex:281, 343 (arXiv:2301.13167v1, thresholds.tex:343).)

But the opening of Theorem 4.1's proof uses the unmodified notation $\mathcal{K}_\kappa$; it does not explicitly substitute one of these altered categories or specify the new arrows. The paper imports methods and notation from the earlier work, so this textual omission is not conclusive evidence that no intended justification exists. It is the precise place where clarification is needed. (arXiv:2301.13167v1, thresholds.tex:269, 359 (arXiv:2301.13167v1, thresholds.tex:359).)

Accordingly, **Proposition 7.2 identifies a concrete problem with the canonical-point reading of the written step on the original category. Its applicability to the modified-category construction asserted in the reply remains unestablished.** Neither the reply alone nor Proposition 7.2 alone resolves that comparison.

There is a further numerical obligation: preserving the categoricity spectrum does not by itself prove preservation of the Löwenheim–Skolem number. If a replacement had LS number $\aleph_1$, applying the displayed threshold formula to it would give

$$\beth_{(2^{\aleph_1})^+} = \Lambda,$$

rather than the original class's threshold $\beth_{(2^{\aleph_0})^+}$. This is a conditional observation, not a claim that Espíndola's intended modification has LS number $\aleph_1$.

Do Theorem 4.1's hypotheses cover the manuscript's class?

Yes, **taking the manuscript's AEC and spectrum claims as stated**.

Issue	Espíndola's Theorem 4.1	Manuscript's claimed class
Class and language	Any AEC; no additional language restriction in the statement	AEC in a countable finitary relational language
LS number	Threshold uses $LS(\mathcal{K})$	$\mathrm{LS}(K) = \aleph_0$
Set theory	Claimed in ZFC; no CH or SCH hypothesis in Theorem 4.1	Construction assumes CH
Amalgamation and joint embedding	Neither assumed	Both fail, according to Proposition 6.2
Tameness	Not assumed	No tameness verification is needed to match the stated hypotheses
Arbitrarily large models	No separate hypothesis stated	Supplied by the claimed categoricity tail
No maximal models	Not assumed	Unnecessary for applying the stated theorem
Categoricity premise	Some $\lambda \geq \beth_{(2^{LS(\mathcal{K})})^+}$	Categorical at $\lambda = \Lambda^+$, indeed throughout $\mu \geq \Lambda$

The AEC and LS assertions are also separately stated in Theorem 3.13, PDF p. 14. The structural failures appear in Proposition 6.2, PDF pp. 27–28. (02-aec.tex:688–715; 05-conclusion.tex:95–134.)

The broader accessible-category results have SCH qualifications, but these are not hypotheses of Theorem 4.1. The introduction expressly claims the AEC result in ZFC, and Theorem 4.1's proof says "we do not need to use amalgamation." Conditions appearing in intermediate lemmas cannot silently be added to its final statement. (arXiv:2301.13167v1, thresholds.tex:249, 267 (arXiv:2301.13167v1, thresholds.tex:267), 359 (arXiv:2301.13167v1, thresholds.tex:359).)

CH supplies a setting in which the purported ZFC theorem must still apply. It is not an exclusion from its scope.

The thresholds and the explicit contradiction.

The manuscript's Theorem 1.1 states:

> "K has at least two nonisomorphic models of cardinality $\beth_{\omega_2} = H(K)$";

and

> "putting $\Lambda = \beth_{(2^{\aleph_1})^+}$, the class K is categorical in every cardinal $\mu \geq \Lambda$."

It explicitly selects $\lambda = \Lambda^+ > H(K)$. (00-introduction.tex:87–99, Theorem 1.1, PDF p. 2.)

Under CH, substituting $\mathrm{LS}(K) = \aleph_0$ into Espíndola's threshold gives exactly

$$\beth_{(2^{\aleph_0})^+} = \beth_{\omega_2}.$$

Applying Theorem 4.1 at $\lambda = \Lambda^+$ would force categoricity at $\lambda' = \beth_{\omega_2}$, directly contradicting Theorem 1.1(ii). Both endpoints in Theorem 4.1 use $\geq$, so there is no endpoint ambiguity.

For comparison with the other cardinal in the objection, the following is a **derived cardinal-arithmetic comparison**, not a quotation:

$$(\beth_{\omega_1})^+ \leq 2^{\beth_{\omega_1}} = \beth_{\omega_1+1} < \beth_{\omega_2} = \beth_{(2^{\aleph_0})^+} < \Lambda \qquad \text{under CH.}$$

The first inequality is Cantor's theorem; the next strict inequality follows from $\omega_1 + 1 < \omega_2$. The final strict inequality follows from $2^{\aleph_1} \geq \aleph_2$, hence $(2^{\aleph_1})^+ > \aleph_2$.

Even without CH,

$$(\beth_{\omega_1})^+ < \beth_{\omega_2} \leq \beth_{(2^{\aleph_0})^+}.$$

Thus $(\beth_{\omega_1})^+$ is strictly below Espíndola's threshold for an AEC with LS number $\aleph_0$. Noncategoricity only at that smaller cardinal would not contradict Theorem 4.1. This arithmetic observation does **not** establish that E1 correctly describes the manuscript's construction.

One source correction matters here. In the supplied files, Proposition 5.7 actually states

$$|I^M| < \Lambda,$$

with $\Lambda = \beth_{(2^{\aleph_1})^+}$, not $|I^M| < \beth_{\delta^+}$. Also, `04-endpoint.tex` is included before `03-categoricity.tex`: they are PDF Sections 4 and 5 respectively. Lemma 5.2 is on p. 19, Proposition 5.7 on p. 24, and Proposition 7.2 on p. 30. These are source-identification findings, not an adjudication of E1. (03-categoricity.tex:440–441; paper.tex:13–16.)

(3) Confidence levels.

Conclusion	Confidence	Reason
Theorem 4.1 contains the point assertion targeted by Section 7	**High**	Explicit text in the proof, confirmed in the PDF
Proposition 7.2 gives the stated obstruction on the original site	**High, conditional on its construction prerequisites**	The dense-cosieve and injectivity argument is explicit and checkable; this is not a certification of the entire manuscript
Proposition 7.2 also defeats Espíndola's intended modified-category construction	**Low**	The intended modification and identification of its point are not specified sufficiently in the supplied reply
Changing morphisms can matter even when the spectrum is preserved	**High**	Density and canonical points depend on arrows; spectrum preservation alone does not determine them
Theorem 4.1's stated hypotheses cover the class claimed by Theorem 1.1	**High**	Direct comparison of the statements; no missing structural hypothesis appears

Conclusion	Confidence	Reason
The thresholds differ as calculated, and the two theorem statements conflict under CH	**High**	Direct substitution and elementary cardinal arithmetic

(4) What I could not determine. I could not identify the exact modified AEC intended in the reply, its morphisms, its LS number, or the comparison functor connecting its model point to the point considered in Proposition 7.2. Consequently, I cannot conclude that Proposition 7.2 identifies a gap in every intended version of Espíndola's argument.

I also did not independently verify the full construction proving Theorem 1.1, resolve E1, or assess the Lean statement and comparator. The conclusions above distinguish a specific obstruction to a proof step from a counterexample to a theorem. The incompatibility of the two theorem statements is established without deciding which one is true.

(5) Concrete questions for a domain expert.

1. **Which category is meant at the opening of Theorem 4.1's proof?** At `thresholds.tex:359`, does $\mathcal{K}_\kappa$ have the original AEC embeddings, Boolean homomorphisms, pure morphisms, or another specified class of arrows?
2. **Where is the modification justified before the point assertion?** Identify the exact definition or lemma supplying the replacement, its accessibility/directed-colimit properties, its LS number, and preservation of the categoricity spectrum at the relevant cardinals.
3. **Which point does the large categorical model define?** Is it the canonical point whose representable values are $\mathrm{Hom}_K(A, H)$? If another point is used, what construction and comparison establish the claim?
4. **What happens to R_A after modification?** Does Lemma 7.1 still supply allowed arrows? Is R_A dense? Does a small-model arrow into H still cover all of B^H? An explicit failure of one of these conditions would locate the boundary of Proposition 7.2.
5. **Does the proposed modification preserve the numerical threshold?** In particular, how does the argument retain $\beth_{(2^{\aleph_0})^+}$ for the original class if its replacement changes the LS number?
6. **For the original site, does the claimed generating family really generate the full dense topology?** Check the first observation in `thresholds.tex:359` against the explicit dense cosieve R_A. This may locate the disputed step more precisely than the later sentence beginning "Clearly."

Claude Opus on all three questions

Independent report on Espíndola's objections E1 and E2 (Opus) Scope: read-only investigation of `paper/` (TeX and PDF), `lean/`, `espindola-src/` and `x-thread.txt`. I did not read the `astra-*.out` files, did not use the network and did not compile Lean. Every mathematical step below was rechecked by hand, not taken from either side.

Numbering note. `paper.tex` inputs the sections in the order 00, 01, 02, **04, 03**, 05, 06 (`paper/build/paper.tex:8-14`). So `04-endpoint.tex` is **Section 4** ("Two models at the prescribed

endpoint", Theorem 4.5) and `03-categoricity.tex` is **Section 5** (Lemma 5.2, Proposition 5.7). CONTEXT.md says 04-endpoint.tex is Section 5; that is incorrect. The PDF confirms the order (`pdftotext`: Section 4 starts on p. 14, Section 5 on p. 19). Pages below are PDF pages.

Label map (PDF number = TeX label, file:line):

PDF	TeX label	location
Theorem 1.1	`thm:main`	00-introduction.tex:87-100, p. 2
Lemma 3.4 (base order has size $\leq \aleph_1$)	`aec:order-size`	02-aec.tex:167-188, p. 9
Lemma 3.5 (size of a basic structure)	`aec:size`	02-aec.tex:190-210, p. 9
Definition 3.7 (objects, $\leq$_K)	`aec:objects`	02-aec.tex:280-293
Lemma 3.12 (LS), Theorem 3.13 (AEC)	`aec:ls`, `aec:class`	02-aec.tex:575-716, pp. 13-14
Lemmas 4.1-4.4, Theorem 4.5	`end:*`	04-endpoint.tex:17-365, pp. 15-18
Lemma 5.1 (collapse)	`col:collapse`	03-categoricity.tex:28-70, p. 19
Lemma 5.2 (no fixed pattern)	`col:no-pattern`	03-categoricity.tex:76-197, pp. 19-20
Lemmas 5.4, 5.5, 5.6	`cat:cardinals`, `cat:finite-partition`, `cat:finite-patterns`	03-categoricity.tex:228-438, pp. 21-23
Proposition 5.7	`cat:bounded`	03-categoricity.tex:440-485, p. 24
Lemma 5.9, Theorem 5.10	`cat:countable-base`, `cat:tail`	03-categoricity.tex:499-627, pp. 24-26
Proof of Thm 1.1, Prop. 6.2		05-conclusion.tex:3-20, 101-124, pp. 26-27
Lemma 7.1, **Proposition 7.2**, §7.3	`bool:extension`, `bool:obstruction`	06-booleanization.tex:15-207, pp. 28-30

Q1 (E1): Lemma 5.2 vs Proposition 5.7

(1) Short answer There is **no mismatch**, and E1 rests on a misreading.

- **Proposition 5.7 does not bound |I^M| by beth_{δ^+}.** Its statement is `If $M\in K$ and $B^M$ is uncountable, then $|I^M|<\Lambda$, with \Lambda=\beth_{\xi^+}, \xi=2^\delta, \delta=\omega_1` (03-categoricity.tex:7-8, 440-442; 00-introduction.tex:94, 193-196). So $\Lambda = \beth_{(2^{\aleph_1})^+}$. Under CH, $\Lambda \geq \beth_{\omega 3} > \beth_{\omega 2} = \beth_{\delta^+}$.
- **Lemma 5.2 states no cardinal bound at all.** It says no sequence of colors (s_j) passes every test. The cardinal |V_δ|^+ = (beth_{ω1})^+ appears only inside its proof by contradiction, as the size of an auxiliary index set X. Any |X| > |V_δ| would do (03-categoricity.tex:93-95).
- **Proposition 5.7 uses only the conclusion of Lemma 5.2.** The threshold Λ comes from the partition-calculus Lemmas 5.4-5.6, which need a strong limit of cofinality > 2^{ℵ1}.

- **Theorem 1.1(ii) does not depend on Lemma 5.2 or Proposition 5.7.** It is proved directly by Theorem 4.5 (05-conclusion.tex:10-11). That construction builds a model with uncountable base and |I| = |V_{ω2}| = beth_{ω2}. E1 does not engage with Section 4.

If Proposition 5.7 really said "|I^M| < beth_{δ^+}", the paper's own model M_u would violate it, because M_u has |I| = beth_{δ^+} and an uncountable base. The paper never makes that claim. Verdict: the proof has no gap at this point, and E1 gives no reason to think Theorem 1.1(ii) is false.

(2) Detailed argument **(a) Espíndola's description of the class.** It is mostly accurate but simplified.

- **Sorts.** There are more than two: I, B, G(B), G_*, the row carriers Z_{k,n} and the column carriers W_k (Def. 3.2, 02-aec.tex:59-128).
- **|B| ≤ ℵ1: correct.** This is Lemma 3.4 (02-aec.tex:167-188), and follows because every bounded interval of B is countable.
- **Size: correct.** Lemma 3.5 gives `\|M\|=\max\bigl(|I^M|,|B^M|,\aleph_0\bigr)` (02-aec.tex:193). Because B is infinite, this equals max(|I|,|B|).
- **Proposition 5.7: misquoted.** E1 says the bound is beth_{δ^+}; the paper says Λ.

(b) Every occurrence of the bounds in question. Found by grep over `paper/build/sections`.

- **|V_δ|^+:** occurs once, at 03-categoricity.tex:94-95: "Choose a linearly ordered set X with |X|>|V_δ|. Such an X can be the initial ordinal of cardinality |V_δ|^+." V_δ also appears at 03:34, 59 (Lemma 5.1: the collapse lands in V_δ) and at 03:192-194 (|X| ≤ |Q| ≤ |V_δ|, the contradiction).
- **(beth_δ)^+:** never written explicitly. By Lemma 4.2, |V_δ| = |V_{ω+ω1}| = beth_{ω1} (04-endpoint.tex:56-91), so |V_δ|^+ = (beth_{ω1})^+.
- **beth_{δ^+} = beth_η = beth_{ω2}:** 04-endpoint.tex:3-6 (η = δ^+ = ω2), 62, 337, 346, 351 (Theorem 4.5); 05-conclusion.tex:7-8, 16; 00-introduction.tex:5, 7, 93, 149-151.
- **H(K):** defined at 00-introduction.tex:21 as `H(K)=h(\LS(K))`, `h(\kappa)=\beth_{(2^\kappa)^+}`. Used at 00:49-50, 58, 66, 93, 98 and 05:7, 16. Under CH and LS = ℵ0, H(K) = beth_{ω2} (05:7-8).
- **Λ:** 00:94, 107, 158, 196; 03:8, 15-17, Lemma 5.4 (03:228-264), Lemma 5.5 (03:267-292), Lemma 5.6 (03:398-433), Proposition 5.7 (03:441-445), Theorem 5.10 (03:603-626); 05:12-18, 30-31, 40, 110, 120; 06:158, 188.

Lemma 5.2 is cited at 01-colors.tex:202, 03:201, 212, 483, 492 and 06:44. **None of these uses |V_δ|^+ as a bound on a model.**

(c) Reconstructing Lemma 5.2. I checked this proof.

1. Suppose (s_j) is a sequence of colors such that T_k((s_{|J|})*J*) *holds for every k. Each s_k decodes to a valid diagram D_k, and D_k↾J = D{|J|}* (03:88-90).
2. For any linearly ordered X, transport D_{|A|} to every finite A ⊆ X. This gives a directed system of quotients Q_A whose maps are injective, preserve tags, and preserve and reflect R (Lemma 2.4, 01-colors.tex:169-196).
3. The direct limit Q is extensional, and R strictly decreases the tags, which lie below ω1 (03:115-178).
4. The roots q_u (u ∈ X) are pairwise distinct by validity condition (iii) (03:180-189).
5. The tagged Mostowski collapse (Lemma 5.1) embeds Q into V_{ω1} (03:190-194). Hence |X| ≤ |V_{ω1}| for every X.

6. Choosing |X| > |V_{ω1}| gives a contradiction.

So the lemma's conclusion is unconditional: **no fixed pattern exists**. The cardinal |V_δ|^+ is a device inside the proof, not an output.

(d) Reconstructing Proposition 5.7. I checked this proof as well.

1. Suppose |I^M| ≥ Λ and B^M is uncountable. Fix a well-order of I^M and a global row-choice system.
2. Color each j-subset of I^M by the whole profile b ↦ c_t(b) ∈ S of its increasing enumeration. Because |S| = ℵ1 under CH and |B^M| ≤ ℵ1, the palette has size ≤ ℵ1^{ℵ1} = 2^{ℵ1} = ξ (03:448-461).
3. Lemma 5.6 (03:398-438) gives profiles p_j and, for each k, a k-set A_k all of whose j-subsets have profile p_j.
   - Its proof keeps an invariant: for each j there are p_j-homogeneous samples of every size below Λ.
   - Pigeonhole is applied to unboundedly many target sizes. This works because ξ < cf(Λ) = ξ^+ (Lemma 5.4, 03:228-265).
   - Each step uses the end-homogeneous (Erdős–Rado-type) estimate, Lemma 5.5 (03:267-391). For the step from m to m+1 it takes q = ξ^ζ and μ = q^+ < Λ. I checked the residual-tree count (03:346-357) and the arity reduction (03:377-390).
4. The tuples t_k have countable exceptional sets E_{t_k} (object condition, Def. 3.7). The union E of these countably many sets is countable, so there is b ∈ B^M ∖ E.
5. Put s_j = p_j(b). Then T_k((s_{|J|})_J) holds for every k, which contradicts Lemma 5.2.

Why the threshold is Λ and not something smaller: b must be chosen **after** the tuples, because the exceptional sets depend on the tuple (Remark 5.8, 03:487-495). So the coloring must use whole profiles (palette 2^{ℵ1}). That in turn needs a strong limit with cofinality above 2^{ℵ1}, which is what Λ = beth_{(2^{ℵ1})+} provides. If one colored by a single fixed coordinate (palette ℵ1), the extracted tuples could all be exceptional at that coordinate.

Nowhere does |V_δ|^+ enter this cardinal computation, and there is no "interchange of successor and beth": the two cardinals involved are (beth_{ω1})^+ and beth_{(2^{ℵ1})+}.

(e) Theorem 1.1(ii) is proved independently in Section 4. I checked every step.

- **Lemma 4.1** (04:17-54) gives f_α: ω1 → ω1 for α < ω2, increasing modulo countable sets. This is the standard diagonal construction and holds in ZFC. E_{γ,α} ⊆ j_α(γ)+1 is countable.
- **Lemma 4.2:** |V_{ω+γ}| = beth_γ, and ω+ω2 = ω2, so |V_{ω2}| = beth_{ω2}.
- **Lemma 4.3:** countable sets D_X ⊆ V_{ω2}, monotone in X, on which membership is extensional. The witness w(a,b) lies in V_{ω2} because V_{ω2} is transitive.
- **Lemma 4.4 (04:212-279):**
  - Tags are f_{rank(·)}(β).
  - Outside the countable set F_X, every membership edge y ∈ x in D_X satisfies f_{rank y}(β) < f_{rank x}(β).
  - The quotient is (D_X, ∈), which is extensional, and the roots x_j are distinct.
  - Restriction to a subtuple gives the subtuple's own diagram, because q depends only on the underlying set.
  - F_{X_J} ⊆ F_X.
  - Hence the full subsequence test passes at every β ∉ F_X.

- **Realization:** B = ω1 ×_lex ℚ is dense, has no endpoints, has countable bounded intervals and has size ℵ1. Lemma 3.8 realizes the colors, so E_t ⊆ {b_β : β ∈ F_X} is countable and M_u ∈ K.
- **Conclusion:** |I| = beth_{ω2} and |B| = ℵ1, while M_c (base ℚ, same |I|) has a countable base. An isomorphism preserves the sort predicate, so M_u ≇ M_c. Both have size beth_{ω2} = H(K).

CH is used only in the coding lemma (|𝔇_k| ≤ 2^{ℵ0}, since ω1^{ℵ0} = 2^{ℵ0}; 01-colors.tex:93-107).

Consequence: by the downward LS property (keep B, choose κ index points), K has ≥ 2 models in **every** κ ∈ [ℵ1, beth_{ω2}]. This includes (beth_{ω1})^+ and beth_{ω2}, not only (beth_{ω1})^+.

(f) A plausible source of E1, and where (beth_{ω1})^+ really does appear. This is my own observation, not in the paper.

Claim: if H ∈ K, b ∈ B^P B^H and H ≤*K P, then* |*I^H*| ≤ |*V*{ω1}| = beth_{ω1}.

Proof sketch: 1. In a strong extension, the full test for every old tuple passes at the new point b, with no exceptions (Def. 3.7, 02:286-292; canonical by Lemma 3.6). 2. Using increasing tuples with respect to a well-order of I^H, the decoded diagrams at b form exactly the compatible directed system used in the proof of Lemma 5.2. 3. So I^H injects into V_{ω1}.

So models with more than beth_{ω1} index points cannot get a new base point by a **strong extension**. This does not bound models with an uncountable base: those allow countable, tuple-dependent exceptional sets, and M_u is built directly, not as such an extension. If Espíndola's intuition is "|V_δ|^+ is where uncountable base must stop", it confuses these two situations. (This claim also matters for Q3; see there.)

(3) Confidence

- **High:** E1 misquotes Proposition 5.7 (it is |I^M| < Λ = beth_{(2^{ℵ1})+}, not beth_{δ^+}). This is a textual fact.
- **High:** Lemma 5.2 states no cardinal bound, and Proposition 5.7 uses only its conclusion. Textual, and the logic is simple.
- **High:** Theorem 1.1(ii) is proved by Theorem 4.5, without Lemma 5.2 or Proposition 5.7 (05-conclusion.tex:10-11).
- **Medium-high:** Theorem 4.5 and Proposition 5.7 are correct. I checked each step myself, and both are short. The Lean development (Q2) formalizes both, which raises confidence once the comparator confirms.
- **Medium:** the claim in (f). It is my own short argument and has not been checked by anyone else.

(4) What I could not determine

- Whether Espíndola has some other argument in mind that he has not written down. His thread text only cites Lemma 5.2's |V_δ|^+.
- I did not re-verify the full AEC axioms (directed unions, smoothness, coherence, LS) line by line beyond a careful read. They looked correct and are not challenged by E1.

(5) Questions for a domain expert

1. Do you agree that Lemma 5.2's statement contains no cardinal, and that |V_δ|^+ is only the size of an auxiliary set in its proof (03-categoricity.tex:93-95)?

2. Is Theorem 4.5 correct? The key checks are Lemma 4.1 (ω2 functions ω1→ω1, increasing mod countable) and Lemma 4.4(ii) (validity outside the countable F_X). If yes, K has two models in every cardinal in [ℵ1, beth_{ω2}] regardless of Section 5.
3. Is the object condition (countable exceptional set per tuple, Def. 3.7) applied the same way in Theorem 4.5 and in Proposition 5.7?
4. Does the claim in (f) (a strong extension adding a base point forces |I| ≤ beth_{ω1}) match the bound Espíndola has in mind?

Q2 (Lean): what `OAI.CHObstruction.main` says and whether it bears on E1

(1) Short answer

(a) The Lean theorem states, assuming 2^{ℵ0} = ℵ1, that there is a countable finitary relational language L and a pair (objects, strong) that:

- satisfies the standard Shelah AEC axioms;
- has least LS number ℵ0;
- satisfies beth_{ω2} = beth_{(2^{ℵ0})+};
- has two non-isomorphic models of cardinality beth_{ω2};
- is categorical (including existence) in every μ ≥ beth_{(2^{ℵ1})+}.

This is Theorem 1.1, essentially verbatim.

(b) I found no vacuity, weakened definition or wrong cardinal. The challenge definitions are character-for-character identical to the ones in the proof module.

(c) Lemma 5.2 ↔ `NoPattern.impossible / Tests.no_pattern`. Its auxiliary index type has cardinality `succ (vonNeumann ω₁).card`, i.e. |V_{ω1}|^+, exactly as in the paper, and its conclusion is `False`, so no cardinal is exported. Proposition 5.7 ↔ `TailBound.index_lt / Data.uncountable_index_bound`. Its conclusion is `#M.I < tailThreshold`, i.e. beth_{(2^{ℵ1})+}.

(d) If the kernel accepts `main` with only `propext`, `Quot.sound` and `Classical.choice`, then two non-isomorphic models of size beth_{ω2} = H(K) exist under CH, as a theorem. E1, as a claim that the models exist only at (beth_{ω1})^+, would then be refuted at the level of the statement. The Lean proof of the two models (`EndpointModels.twoModels`) does not even use the Lemma 5.2 / Proposition 5.7 analogues.

(2) Detailed argument (a) The statement (`lean/CHObstruction.lean`; definitions identical to `lean/OAI/ModelTheory/Categoricity/Model.lean:26-107`; I diffed them programmatically and they are identical).

Structures and maps. - `RelModel L`: a carrier that is a `ZFSet.{u}`, with relations `rel n r : (Fin n → carrier) → Prop`, where `L : ℕ → Type` gives the n-ary symbols. `Countable (Σ n, L n)` means a countable finitary relational language. - `IsSubstructure`: carrier inclusion, with relations agreeing on the smaller carrier (induced substructure). - `Iso`: a bijection preserving and reflecting all relations.

AEC axioms (`IsAEC`): - strong ⇒ both objects; reflexive, antisymmetric, transitive; strong ⇒ substructure; - the class of objects is closed under isomorphism; - `iso_strong`: if M ≤ N, and e: M≅M′ and f: N≅N′ agree on M, then M′ ≤ N′ (Shelah's invariance axiom); - `coherence`; - `chain_union`:

for every nonzero ordinal γ and every $\leq$-chain indexed by γ, there is an object U that is the union (each member is a substructure and every element lies in some member), each member is $\leq$ U, and smoothness (if all members are $\leq$ N then U $\leq$ N); - `downward`: some LS bound exists.

LS number. `LSBound θ`: $\aleph_0 \leq \theta$, $|L| \leq \theta$, and for every object M and every $A \subseteq M$ there is $N \leq M$ with $A \subseteq N$ and $\|N\| \leq |A| + \theta$. `HasLSNumber ℵ₀` means $\aleph_0$ is such a bound and is $\leq$ every bound.

Categoricity and two models. - `Categorical μ`: there exists an object of cardinality μ, and any two objects of cardinality μ are isomorphic. - `TwoModels μ`: two non-isomorphic objects of cardinality μ.

Cardinals. - `CH`: `2^ℵ₀ = ℵ_ 1`. - `hanf κ = beth (succ (2^κ)).ord`, i.e. beth_{(2^{κ})+}, with `succ` the cardinal successor and `.ord` the initial ordinal. - `endpoint = beth (ω_ 2)` = beth_{$\omega 2$}. - `tailThreshold = beth (succ (2^ℵ_1)).ord` = beth_{($2^{\{\aleph 1\}}$)+} = Λ. - Mathlib's `beth 0 = ℵ0`. The proof of `EndpointDiagrams.cardinal` goes through `ZFSet.card_vonNeumann` and `preBeth`, so the kernel itself checks that |V_{$\omega 2$}| = `endpoint`.

Conclusion of `main`. `∃ L countable, ∃ K, K.IsAEC ∧ K.HasLSNumber ℵ₀ ∧ endpoint = hanf ℵ₀ ∧ K.TwoModels endpoint ∧ ∀ μ ≥ tailThreshold, K.Categorical μ`.

`Main.lean:253-264` also derives `prescribed_transfer_fails`: K is categorical in `succ tailThreshold` > `hanf ℵ₀`, but not categorical in `hanf ℵ₀`. This is exactly the negation of the prescribed-threshold transfer for this K.

(b) Faithfulness. Theorem 1.1 asks for: - an AEC in a countable finitary relational language ✓ - LS(K) = $\aleph_0$ ✓ (least, and the language-size clause is included) - two non-isomorphic models of size beth_{$\omega 2$} = H(K) ✓ - categoricity in every $\mu \geq \Lambda$ = beth_{($2^{\{\aleph 1\}}$)+}, with existence ✓ - CH ✓

Checks against vacuity: - `TwoModels` and `Categorical` both assert existence. Existence on the tail is proved via `Existence.models` (Main.lean:79-83). - `HasLSNumber` asserts a real LS bound. The AEC axioms are the standard ones (no axiom is missing; chains are ordinal-indexed, which is the usual formulation). - The proof modules contain no `sorry`, `axiom`, `native_decide`, `implemented_by`, `extern`, `unsafe` or `partial` (grep over `lean/OAI`). - `CHObstruction.json` permits only `propext`, `Quot.sound` and `Classical.choice`.

Minor representation choices, none of which changes the mathematics: - Carriers are `ZFSets`, and every cardinal in `Cardinal.{u}` is attained. - The class is presented via `Data` records with canonical torsor coordinates (`AEC.lean:64-160`): `objects R := ∃ P : Presentation R, P.data.IsObject hCH`. This is the paper's class up to isomorphism.

The Lean statement does not formalize Theorem 1.1's "in particular" remark beyond `prescribed_transfer_fails`, nor Corollary 6.1 or Section 7. `240.md` says this explicitly.

(c) The counterparts of Lemma 5.2 and Proposition 5.7.

Lemma 5.2 ↔ `NoPattern.impossible : False` (`Diagrams.lean:285-296`), wrapped as `Tests.no_pattern` (`Colors.lean:173-207`). Its proof: - sets `V := ZFSet.vonNeumann (ω_ 1)` and `J := (succ V.card).ord.ToType`, an index set of cardinality |V_{$\omega 1$}|^+; - builds the direct limit of the stage quotients and embeds its roots into `vonNeumann ω₁` by the tagged collapse (`Collapse.lean:64-70, 312-330`; the bound comes from `rank_le`); - concludes `succ V.card ≤ V.card`, a contradiction.

So the Lean proof makes the same use of |V_δ|^+ as the paper, and again the exported statement contains no cardinal.

Proposition 5.7 ↔ `TailBound.index_lt` (`TailBound.lean:90-122`):

```
theorem index_lt (hB : #B ≤ ℵ_ 1) (hunc : ¬ Countable B) (c : Tuple I → B →
↪  Palette)
    (hbad : ∀ t, {b | ¬ Test hCH (fun t => c t b) t}.Countable) : #I < tailThreshold
```

- It is used as `Data.uncountable_index_bound` (`Arrays.lean:152-154`) and in `Categoricity.countable` (`Main.lean:103-108`).
- It calls `FinitePatterns.exists_patterns tailThreshold ξ …` with `ξ = 2^ℵ_1` (`Partition.lean:267`), together with `strongLimit` (Λ is a strong limit), `palette_lt_cofinality` ($\xi < \mathrm{cf}\,\Lambda$) and `profile_cardinal` ($|B \to \mathrm{Palette}| \le \xi$).
- It then picks b outside the countable union of exceptional sets and applies `Tests.no_pattern`.

The bound actually used is beth_{($2^{\{\aleph 1\}}$)+}. It is neither beth_{ω2} nor (beth_{ω1})^+.

Theorem 4.5 ↔ `EndpointModels.twoModels` (`Main.lean:209-227`): - I is `Shrink (vonNeumann ω₂)` with `#D = endpoint` (`EndpointDiagrams.lean:168-176`); B is `ω₁ ×ₗ ℚ` with `#B = ℵ_1`. - The object condition comes from `countable_failures` (EndpointDiagrams.lean:164), via `Domination.countable_exceptions` (the analogue of Lemma 4.1). - Non-isomorphism comes from the base's cardinality. - This proof does not use `index_lt` or `no_pattern`.

(d) What a kernel-checked proof implies for E1. The formal statement says that two non-isomorphic models exist at exactly beth_{ω2} = hanf $\aleph_0$. So the assertion "they exist only at (beth_δ)^+, not at beth_{δ^+}" is contradicted by a checked theorem whose definitions are standard and match the paper.

E1 could at most point to a mistake in the prose that does not affect the theorem, and I found none. Espíndola's remark that it is "formalizing something else" is not supported. The only room left is in the definitions, and those are the standard AEC notions, the standard Mathlib beth, and CH in its usual form.

Caveats: the Lean meta-theory is stronger than ZFC (it includes universes), but nothing here uses large cardinals beyond the universe hierarchy. Also, acceptance of the formal theorem does not, by itself, certify the TeX proofs.

(3) Confidence

- **High:** the reading of the statement and of the definitions; that they are identical to the solution's definitions; and which cardinal each Lean lemma uses. I read the code directly.
- **Medium-high:** no hidden vacuity. Every clause asserts existence and the axioms are standard, but I did not compile anything.
- **Conditional:** "the Lean result refutes E1" holds only if the comparator accepts the proof with the three standard axioms. That result is not available to me.

(4) What I could not determine

- Whether the code compiles and the kernel accepts it (no build here; the comparator is pending).
- The Mathlib version. It is not pinned in `lean/`. Semantics of `beth`, `ω_` and `ℵ_` depend on it; they match current Mathlib (beth 0 = $\aleph_0$, `ω_ o = (ℵ_ o).ord`).
- I did not read the ~3,000 lines of `Arrays*`, `Subarrays`, `DirectLimits` and `AEC.lean` that verify the AEC axioms in detail. Those only affect whether the proof goes through, not what the statement means.

(5) Questions for a domain expert or Lean reviewer

1. Does the comparator accept `OAI.CHObstruction.main` against the challenge file, with only `propext`, `Quot.sound` and `Classical.choice`?
2. Is `IsAEC` (especially `iso_strong`, `coherence` and `chain_union` with smoothness) an acceptable rendering of Shelah's AEC axioms?
3. In the pinned Mathlib, does `Cardinal.beth` satisfy beth $0 = \aleph_0$ and the usual successor and limit clauses?

Q3 (E2): Espíndola's Theorem 4.1 vs the manuscript's Theorem 1.1 and Proposition 7.2

(1) Short answer

(b) first, because it is the decisive point. **Espíndola's Theorem 4.1, as stated, covers the manuscript's AEC.**

- The statement is: `Let $\mathcal{K}$ be an AEC. If $\mathcal{K}$ is categorical in some $\lambda \geq \beth_{(2^{LS(\mathcal{K})})^+}$, then $\mathcal{K}$ is $\lambda'$-categorical for every $\lambda' \geq \beth_{(2^{LS(\mathcal{K})})^+}$` (thresholds.tex:354-356).
- It has no hypotheses on the language, and no amalgamation, JEP, tameness, no-maximal-models or set-theoretic hypotheses. It is claimed in ZFC.
- For the manuscript's K: LS(K) = $\aleph_0$, so the threshold is beth_{$(2^{\{\aleph 0\}})$+}, which is beth_{$\omega 2$} under CH. K is categorical in $\lambda = \Lambda^+ >$ beth_{$\omega 2$}. Theorem 4.1 therefore asserts categoricity at beth_{$\omega 2$}, and Theorem 1.1(ii) asserts two models there.
- **Theorem 1.1 (under CH) and Theorem 4.1 (claimed in ZFC) cannot both be true** unless ZFC is inconsistent: by Gödel, Con(ZFC) ⇒ Con(ZFC+CH), and the two theorems together give ZFC ⊢ ¬CH.

On "same models, different morphisms": passing to such a category does not change which models exist or which are isomorphic, so it cannot make K categorical at beth_{$\omega 2$}. It may change the site used in the proof. That is a statement about where the proof breaks, not about whether the two theorems conflict.

(a) Prop 7.2 targets a step that appears in the written proof of Theorem 4.1 in arXiv v1:

- the proof works with "the double negation subtopos of `\Sets^{\mathcal{K}_{\kappa}}`", κ = LS(K), that is, presheaves on the countable models with K-morphisms;
- it then asserts "Clearly the model of size λ must be one of its points" (thresholds.tex:359).

The v1 text does not mention modified morphisms at this step. Prop 7.2 shows that the canonical point of any large categorical model of K does not factor through the full ¬¬-subtopos of that site. The paper explicitly does not exclude other points or other topologies (06-booleanization.tex:200-207). So 7.2 identifies a step in the proof as written in v1, conditional on "points" meaning canonical points. Whether it hits the argument Espíndola now says he uses (a modified category) cannot be decided from the source.

(2) Detailed argument (b) Hypotheses and thresholds.

Espíndola's notion of AEC. In 1906.09169, Def. at scc3.tex:264-278, an AEC is a category equivalent to an accessible category with directed colimits and monomorphisms, having an iso-full, nearly full embedding into a finitely accessible category. He asserts the link to Shelah's definition and the agreement of internal size with cardinality (scc3.tex:280). By Beke–Rosický, every Shelah AEC is such a category (an expert should confirm this is the reading he intends). The manuscript's K is a Shelah AEC with LS(K) = ℵ0 (Theorem 3.13). Section 7 also argues that internal size equals cardinality for K (06:114-130).

Hypotheses of Theorem 4.1. None beyond "AEC" and categoricity in some $\lambda \geq \beth_{(2^{\{LS(K)\}})+}$. No CH or ¬CH, no amalgamation, no tameness, no arbitrarily large models (categoricity at λ already gives a large model). Its proof is said to need no amalgamation ("we do not need to use amalgamation", thresholds.tex:359).

K satisfies the hypotheses. By Theorem 5.10, K is categorical in every $\mu \geq \Lambda$, in particular in $\lambda = \Lambda^+ \geq \beth_{(2^{\{\aleph 0\}})+}$.

Conclusion of 4.1 for K. K is categorical at $\beth_{(2^{\{\aleph 0\}})+}$, which is $\beth_{\omega 2}$ under CH. **This contradicts Theorem 4.5 / Theorem 1.1(ii).**

Thresholds compared (ZFC unless stated):

cardinal	where it appears	relation
$(\beth_{\omega 1})^+ = \lvert V_\delta \rvert^+$	Lemma 5.2's auxiliary set; Espíndola's proposed failure point	$< \beth_{\omega 1+1} \leq \beth_{\omega 2}$
$\beth_{(2^{\{\aleph 0\}})+}$	Thm 4.1 threshold for LS = ℵ0; H(K)	$\geq \beth_{\omega 2}$; **= $\beth_{\omega 2}$ under CH**
$\beth_{\omega 2} = \beth_{\delta^+}$	Thm 1.1(ii): two models	= H(K) under CH
$\Lambda = \beth_{(2^{\{\aleph 1\}})+}$	Thm 1.1(iii) tail; Prop 5.7 bound	$> \beth_{\omega 2}$ (since $2^{\{\aleph 1\}} \geq \aleph 2$)

- If the only failure were at $(\beth_{\omega 1})^+$, Theorem 4.1 would indeed be untouched, so Espíndola's conditional is correct.
- But the manuscript places a failure at $\beth_{\omega 2}$ = the 4.1 threshold itself.
- In fact K is non-categorical in every $\kappa \in [\aleph 1, \beth_{\omega 2}]$ (Q1(e)).

Different morphisms do not rescue the theorem. Suppose K′ has the same models as K and its morphisms are still embeddings. Then K′-isomorphisms are τ-isomorphisms, and K′ has the same categoricity spectrum by cardinality (he says so himself). So K′ is again categorical at Λ^+ and not at $\beth_{\omega 2}$.

- If K′ is an AEC with LS(K′) = ℵ0, Theorem 4.1 applied to K′ is contradicted in the same way.
- If LS(K′) > ℵ0, applying 4.1 to K′ says nothing about $\beth_{\omega 2}$. But then a proof that goes through K′ does not prove 4.1 for K as stated.

Either way the theorem-level conflict remains.

Internal discrepancy in 2301.13167 (an expert should look at this): - Theorem 4.1 and the introduction use $\beth_{(2^{\{LS(K)\}})+}$ (thresholds.tex:249, 355). - The abstract theorem and Theorem 6.1 bound the start of an eventually-categorical spectrum by $\beth_{(2^{\kappa})+}$, with κ the accessibility index (thresholds.tex:231-236, 382-389). - For an AEC with LS = ℵ0, the natural κ is ℵ1, which gives $\beth_{(2^{\{\aleph 1\}})+} = \Lambda$, exactly the manuscript's tail. - With κ = ℵ1, Theorem 6.1 is not directly

contradicted by Theorem 1.1 as stated. It would be contradicted only if K had uncountable-base models with |I| cofinal in Λ, which the manuscript does not determine. - Theorem 4.1, stated with LS(K), is directly contradicted. The proof of 6.1 itself switches between "2^κ-tame" and "2^{κ+}-tame" (thresholds.tex:359 vs 394).

(a) Does Prop 7.2 hit a step the proof uses?

The step. The proof of 4.1 begins: "Consider the double negation subtopos of `\Sets^{\mathcal{K}_{\kappa}}`, corresponding to the dense topology, when `\kappa=LS(\mathcal{K})`." It then says "Clearly the model of size λ must be one of its points" (thresholds.tex:359). By 1906.09169's notation, scc3.tex:334, $\mathcal{K}$_κ is the full subcategory of objects of internal size κ, i.e. the countable models with K-morphisms. That is the manuscript's site C_0 (06:92-130).

What Prop 7.2 proves. I checked the proof.

- Lemma 7.1 (06:15-88):
  - every countable model has a strong extension adding a base point;
  - the colors at the new point come from a diagram on ω, tagged by the natural numbers themselves, with ordinal membership;
  - these colors are tuple-dependent, so they do not contradict Lemma 5.2.
- The cosieve R_A = {r : r adds a base point} is therefore ¬¬-dense (06:161-174).
- Let H be a model with countable base, and A ≤ H countable with B^A ⊇ B^H (images). Then a: A → H does not factor through any r ∈ R_A, because h would have to send a new base point into B^H = a[B^A], contradicting injectivity (06:176-181).
- So p_H^(*R_A*) → *p_H*^(y_A) is not surjective. Hence the canonical point p_H (p_H^*(y_A) = Hom(A,H)) does not factor through Sh(C_0^op, J_¬¬) (06:183-190).
- All models of size ≥ Λ have countable base (Prop 5.7), so this applies to every large categorical model.

The point formula is the standard one for presheaves on the opposite of C_0 (flat functors C_0^op → Set). I found no error in this argument.

Its limits, which the paper itself acknowledges (06:195-207): - it rules out only the canonical factorization; - it does not classify the points of the Booleanization; - it says nothing about other sites, morphism classes or topologies.

Tension inside Espíndola's own description. He describes the dense topology as "generated by those covers {M → N_i} ... such that any M → N, for an arbitrarily large model N, can be amalgamated with one M → N_j" (thresholds.tex:359). By the claim in Q1(f), a large H (|I^H| > beth_{ω1}, countable base) has no strong extension adding a base point. So A → H cannot be amalgamated over A with any member of R_A, and R_A is **not** one of his generating covers.

That leaves two options: - his topology is smaller than ¬¬, and then his step "This guarantees that the sheaf topos is Boolean (whence, it is the double negation subtopos)" needs checking, since for R_A one has ¬R_A = ∅ by Lemma 7.1, so R_A ∨ ¬R_A = R_A is not a cover of his kind; or - it is the full ¬¬-topology, and then Prop 7.2 blocks the canonical point.

This is a concrete place where an expert can see the two descriptions diverge.

Espíndola's reply ("an axiomatization of a category ... equivalent to a modified AEC (same models, different morphisms)"). - In v1 the only morphism changes I found are elsewhere: in Lemma 2.1 / §3 (categories $\mathcal{K}$^B of "κ-Boolean homomorphisms" and $\mathcal{K}$^r of Reg_⊥-models, thresholds.tex:281,

320-326), and in the Baldwin–Boney–Vasey expansion, which he calls "an isomorphic AEC" (thresholds.tex:304). - The first sentence of the proof of 4.1 uses `\Sets^{\mathcal{K}_{\kappa}}`. - If the intended site is $\mathcal{K}$^B_κ or similar, Lemma 7.1's extensions may not be morphisms there, and Prop 7.2 would not apply as stated.

(3) Confidence

- **High:** Theorem 4.1 as stated (thresholds.tex:354-356) has hypotheses that K satisfies, and its conclusion contradicts Theorem 1.1(ii). So Theorem 4.1 and Theorem 1.1 cannot both be true. This is textual plus elementary cardinal arithmetic. It depends only on Espíndola's "AEC" including Shelah AECs, which he asserts (scc3.tex:280).
- **High:** changing morphisms while keeping the models and isomorphisms cannot make K categorical at beth_{ω2}.
- **Medium-high:** Prop 7.2 is correct as stated (canonical point, full ¬¬ topology on countable models with K-morphisms).
- **Medium:** Prop 7.2 hits the step Espíndola's written v1 proof uses. The step is stated tersely, and "points" is not defined at that place.
- **Medium:** the observation that R_A is not among his amalgamation-generated covers. It relies on my own claim in Q1(f).

(4) What I could not determine

- Which category and topology Espíndola actually intends in the first step of 4.1 ($\mathcal{K}$_κ with K-morphisms, or a modified one), and whether "points" means canonical points.
- Whether a later arXiv version of 2301.13167 changed Theorem 4.1 or its proof. I had only v1.
- Whether the accessibility index κ in Theorem 6.1 is meant to be LS(K) or LS(K)^+ for AECs. This decides whether 6.1, unlike 4.1, is compatible with Theorem 1.1.
- Whether K has uncountable-base models of size cofinal in Λ (relevant only to Theorem 6.1).

(5) Questions for a domain expert

1. Does Theorem 4.1 of arXiv:2301.13167, applied to the manuscript's K (LS = ℵ0, categorical in Λ^+, under CH), assert categoricity at beth_{ω2}? If yes, at least one of the two papers is wrong. Which proof do you find more convincing at the decisive step?
2. In the proof of Theorem 4.1, which category is $\mathcal{K}$_κ, and which morphisms does it use? Is "the model of size λ must be one of its points" meant as the canonical point Hom(−, M) restricted to countable models? If so, does Proposition 7.2 refute that step for this K?
3. Is the topology "generated by covers amalgamable with arbitrarily large models" equal to the ¬¬-topology on $\mathcal{K}$_κ when amalgamation fails, as it does for this K (Prop. 6.2)? Specifically, is the cosieve R_A of Prop. 7.2 covering for it?
4. If the proof uses a modified category with the same models, what is its LS number? Does Theorem 4.1 for K then follow from 4.1 for the modified class?
5. For AECs, is the κ in Theorem 6.1 LS(K) or LS(K)^+? With κ = LS(K)^+, the bound beth_{$(2^{\kappa})^+$} equals the manuscript's Λ.

Appendix E. Round-2 reports

Astra, blind referee

The supplied proof does **not establish Theorem 4.1**. Several essential steps appear wrong, particularly the identification of points and slices after changing morphisms. The explicit cardinal threshold also rests on an unstated omitting-types argument.

Write **A** for arXiv:2301.13167v1, 2301.13167/x/thresholds.tex, and **B** for arXiv:1906.09169v7, 1906.09169/x/scc3.tex. Citations give TeX lines and the corresponding paper's PDF pages; printed and PDF page numbers agree.

The architecture is: set $\kappa = LS(\mathcal{K})$, $H = \beth_{(2^\kappa)^+}$, and assume categoricity at $\lambda \geq H$. Booleanize the presheaf topos on κ-sized models; use the categorical model to select an atomic component. Its points should form a class categorical at κ^+ and λ. Lemma 2.1 should make this an AEC; Theorems 3.1–3.2 should supply tameness and further categoricity; B should extend categoricity everywhere above κ^+. Finally, a Morley-style argument should put every sufficiently large original model into this component, with cutoff below H.

In the table, C is a skeleton of $\mathcal{K}_\kappa$, with **original strong embeddings**; $E = \mathbf{Set}^C$ is the presheaf topos on C^{op}, and $D = \mathrm{Sh}(C^{op}, J_{\neg\neg})$. A κ^+-point has inverse image preserving all limits of size at most κ.

Step and citation	Claim and assessment	Rating
1. Presheaf point. A:359, p.7; B:494–504, p.14	A model M gives $p^*_M(F) = \mathrm{colim}_{A \leq_{\mathcal{K}} M,\ \lvert A\rvert=\kappa} F(A)$. The indexing poset is κ^+-directed by downward LS and coherence, so this is a κ^+-point of E.	**follows**
2. Dense covers. A:359, p.7	On C^{op}, density means every small extension of the source admits a further extension meeting the sieve. Replacing this by amalgamability with arbitrarily large extensions requires proof. Moreover, $[N,-] \to [M,-]$ induced by an embedding $M \to N$ is generally **not monic**: extensions of a map need not be unique. Existential subobjects are its images. The displayed argument does not establish the proposed topology's equality with $J_{\neg\neg}$.	**appears wrong**

Step and citation	Claim and assessment	Rating
3. The categorical model factors through D. A:359, p.7; B:629–636, pp.18–19; B:669–688, p.20	"Clearly" conceals the requirement that every dense covering family lift every map $A \to M_\lambda$ through some member **inside** M_λ. Compatibility in an extension is insufficient. B Lemma 6.4 claims λ-closedness, but depends on the problematic atomicity arguments below. B Theorem 7.1 identifies dense-topology points with saturated models **assuming amalgamation**. It does not establish the claimed identification without amalgamation.	**unclear**
4. Isolation and slicing. A:359, p.7	Boolean alone does not imply atomic. Nevertheless, **conditional on a point p of D**, its completely prime filter on the complete Boolean algebra $\mathrm{Sub}_D(1)$ is principal at an atom S. Thus p factors through D/S, which is two-valued and Boolean. This repairs isolation of the **subterminal truth type**, not isolation of tuple/Galois types.	**follows with a fixable gap**
5. Equivalence versus isomorphism. A:359, p.7; B:580–608, pp.16–18	B Theorem 6.1 concerns a specified Boolean fragment, cardinal-arithmetic assumptions, and models of the matching size; Corollary 6.2 requires separability and directed colimits of points. A switches between κ^+-points, $L_{\infty,\kappa}$-equivalence, and "κ-points." $L_{\infty,\kappa}$-equivalence alone does not imply isomorphism at size κ^+. Correct indices and a justified back-and-forth argument are needed.	**unclear**

Step and citation	Claim and assessment	Rating
6. Saturated model and component. A:359, p.7	Categoricity at λ is asserted to provide the stability needed for a κ^+-saturated model of size κ^+, without specifying the stability cardinals or proving the needed hypotheses. Even existence would not show that every saturated model belongs to the particular component D/S: that requires agreement with the chosen point on subterminals.	**unclear**
7. A small axiomatized class. A:359, p.7; B:367–410, pp.8–10	The points of D/S are claimed to define an L_{κ^{++},κ^+}-axiomatized κ^+-AEC. One must specify its morphisms, underlying-set functor, accessibility and LS bound. B's enough-points theorem requires a suitably small site/fragment and $\alpha^{<\alpha} = \alpha$. Here C can have 2^κ objects and many more sieves; the required small presentation is not supplied.	**unclear**
8. Applying Lemma 2.1. A:276–298, pp.3–4; A:359, p.7	Lemma 2.1 assumes amalgamation and stability at every relevant $\rho < r$, for r-saturation. Applied here, $r = \kappa^+$. Those hypotheses are not verified. The subsequent assertion that closed points become saturated upon selecting S does not prove the missing extension property. B:657–688 explicitly defines saturation by injectivity against small embeddings and uses amalgamation.	**appears wrong**

Step and citation	Claim and assessment	Rating
9. Lemma 2.1's slice calculation. A:281–294, p.3; B:786–796, pp.24–25	Let $C_0 = \mathcal{K}_{\geq r,<\eta}$, and C_0^B retain only r-Boolean morphisms. Restriction sends $\mathrm{Hom}_{C_0}(M,-)$ to that same functor restricted to C_0^B, **not** to $\mathrm{Hom}_{C_0^B}(M,-)$. Consequently the pullback slice has objects involving all original arrows out of M, not merely Boolean arrows. The claimed Boolean coslice and subsequent limit/Yoneda argument therefore do not follow.	**appears wrong**
10. Lemma 2.1's types argument. A:296–298, pp.3–4	Stability of Galois types is transferred to Boolean syntactic types, then an omitting-types theorem supposedly makes subobject lattices atomic and Boolean. Neither the comparison nor the implication is proved. Likewise, a saturated P need not be universal across unrelated components. The ensuing equality of subobject lattices does not follow from two-valuedness. Ordinary lattice atomicity also does not by itself imply Booleanness.	**appears wrong**
11. Presheaf reconstruction used throughout. A:322, p.5; B Theorem 4.1, B:487–537, pp.13–15	B identifies a β-classifying topos with presheaves on β-presentable models, under accessibility and cardinal assumptions. Its proof treats a conservative inverse image $E_0 \to \mathbf{Set}^I$ as though it were a full inclusion, then descends G through it from object presentations. Conservativity does not provide this descent of objects, arrows and naturality. This is a substantive gap in the cited proof.	**unclear**

Step and citation	Claim and assessment	Rating
12. Omitting types. B Theorems 5.1–5.2, B:544–572, pp.15–16	The statements assume regular α, a Boolean α-fragment, directed colimits and GCH. The proof adds omission covers, then claims their density survives passage to the larger-fragment presheaf topos. This is false in general: a small model realizing the type can prevent that cover from being dense at its realizing tuple. See the concrete check below.	**appears wrong**
13. Booleanization and closedness. B Corollary 6.2/Lemma 6.4, B:611–636, pp.18–19	B:614 concludes that a surjection is an embedding, hence an equivalence, because source and target have equivalent double-negation subtoposes. Equivalent Booleanizations do **not** determine the original toposes or make such a morphism an embedding. Lemma 6.4 reuses this argument, including to claim directed colimits for Booleanized models. Thus it cannot currently justify Step 3.	**appears wrong**
14. Regular Morleyization. A:304–326, pp.4–5; B:720–740, pp.21–22	Adding predicates for negations produces a larger regular-model category $\mathscr{K}^r$. Disjointness $R \wedge S \vdash \bot$ does not make S the complement of R; excluded middle is added later. The disjunction-elimination equivalence is classical, not generally intuitionistic. Preservation of pseudocomplements by evaluation relies on reconstruction and compactness arguments whose precise applicability is not established here.	**unclear**

Step and citation	Claim and assessment	Rating
15. Theorem 3.1: monster and amalgamation. A:328–333, p.5; B Theorem 8.3, B:744–749, pp.22–23	A requires amalgamation in $\mathcal{K}_{\geq(2^\kappa)^+,<\lambda}$. B Theorem 8.3, from categoricity at $a < b$, concludes amalgamation **above** b. Taking $b = \lambda$ gives the wrong interval. B Theorem 8.2 additionally requires a strongly compact cardinal. Neither supplies the stated monster-model setup.	**appears wrong**
16. Theorem 3.1: types and injectivity. A:333, p.5; B:749, p.23	The asserted Galois/syntactic-type correspondence lacks a located supporting statement in B. Compactness gives a model in $\mathcal{K}^r$; it does not automatically give an original $\mathcal{K}$-model. Injectivity of M_λ against pure embeddings in this larger category, and the required size bounds for the constructed models, are merely asserted by analogy with B's proof.	**unclear**
17. Theorem 3.2. A:338–345, p.6	With $\xi = \beth_\omega(\kappa)$, the proof changes to regular models with ξ-Boolean morphisms, seeking directed bounds and a ξ^+-point of their Booleanization. Nondegeneracy, the regular quotient presentation and the saturation/injectivity identification require verification. Its proposed 3-amalgamation proof says the pushout theory is consistent because amalgamation holds; ordinary amalgamation does not ensure consistency of an entire 3-amalgamation diagram. Weak initiality alone does not ensure the chosen maps are the required pure/Boolean embeddings.	**appears wrong**

Step and citation	Claim and assessment	Rating
18. Applying Theorems 3.1–3.2. A:329,339,359, pp.5–7	The new class is categorical only at κ^+ and λ. Theorem 3.1 therefore yields at best $(2^{\kappa^+}, < \lambda)$-tameness, not global 2^κ-tameness. Theorem 3.2 requires categoricity throughout $[\kappa^+, \beth_\omega(\kappa^+))$, which has not been obtained. Neither statement supplies the asserted proper class of categoricity cardinals from the available premises.	**appears wrong**
19. Transfer from B. A:359, p.7; B:759–818, pp.23–26; B:852–857, p.27	B Theorem 9.1 assumes amalgamation, GCH and categoricity above an unspecified categoricity Hanf number. Its stronger downward argument uses the defective slice calculation, proper Boolean extensions and transfinite Kripke constructions. B Lemma 6.6 handles limits using amalgamation and closedness; Theorem 9.3/Corollary 9.6 provide only an unspecified eventual threshold. None directly establishes categoricity from κ^+ here.	**unclear**
20. Removing GCH. B:866–904, pp.27–30	Small-distributive forcing can preserve small coded models, but the proof also needs preservation of definability, transformations and categoricity. At B:873, sets of long tuples need not have the underlying model's size; at B:885, individually old components do not make an entire natural transformation old. The later thinning of the completeness construction does not establish all claimed conservativity properties.	**unclear**

Step and citation	Claim and assessment	Rating
21. The explicit threshold. A:359 and footnote, p.7	The final "Morley's omitting types theorem for AECs" is neither stated nor proved locally in the needed form. B's Theorem 5.1 gives omission of specified nonisolated syntactic types, not a uniform cutoff $< H$ for nonsaturation. The omitted argument must code failure of saturation/component membership with controlled vocabulary and types, and track LS through all expansions. This is where the **specific beth bound** must be proved.	**unclear**
22. Final assembly. A:359, p.7	If the selected class is categorical at every cardinal above κ^+, and every original model of size at least some $\nu < H$ belongs to that class with the same isomorphisms and cardinality, the stated conclusion follows. Merely proving saturation would still require showing membership in the selected component.	**follows**, conditional on those unproved premises

The most doubtful steps, ranked, are:

1. **Changing morphisms inside slices — Step 9.** Check the inverse image of the representable explicitly. The two Hom-functors differ exactly on embeddings that are not Boolean-preserving. Repair requires a new argument, or proving those embeddings already Boolean without using the desired conclusion.

2. **Transporting omission-cover density — Step 12.** Take a first-order theory with countably many disjoint infinite unary predicates P_n, allowing elements outside their union. The type $\{\neg P_n(x) : n < \omega\}$ is nonisolated; the family $P_n(x)$ is dense in the Boolean syntactic site. In a countable model with an element outside every P_n, elementary embeddings preserve that fact. The corresponding family is therefore not dense above that tuple in the small-model presheaf topos. An expert should check B:572 against this example.

3. **The initial point and saturation identifications — Steps 2–8.** Write out the covering-lifting condition for M_λ, prove it without amalgamation, and distinguish closedness from injectivity against all small embeddings. Then establish that the required small saturated model lies in the *same* component.

4. **The tameness/categoricity bootstrap — Steps 15–19.** Supply the missing below-λ amalgamation theorem, the initial categoricity interval, and a transfer theorem with verified hypotheses. Correct 2^{κ} versus 2^{κ^+}; these are not interchangeable in ZFC.

5. **The bound H — Step 21.** State and prove the precise uniform omission/coding lemma, including parameter sizes and exclusion of other components. An unspecified eventual-categoricity theorem cannot supply this numerical bound.

I could not assess the externally cited regular-logic compactness proofs, the proposed adaptation of Morley's method, or other external references: they are not supplied in the permitted folder. I found no local theorem proving the stability assertion or the cited Galois/syntactic-type correspondence in the required generality. All inspection was read-only and confined to the two supplied papers; no network or outside literature was used.

Astra, test case

Short answer. On the category specified in the question, the first false claim is the "first observation" in arXiv:2301.13167v1, thresholds.tex:359: the proposed families involving amalgamation with arbitrarily large models do **not** generate the full dense topology. The subsequent assertion that the large categorical model supplies a point of the Booleanization is also false.

The argument below strengthens JUDGE-REPORT.md by addressing **generation of the topology**, rather than merely showing that R_A is absent from the proposed generators. Confidence is **high** for the canonical-point obstruction and applicability of the theorem; **medium-high** for the additional generator argument and morphism analysis, which are mathematical deductions here, not Lean-certified results.

I used only local materials, without modifying files. Section and theorem numbers follow the PDFs.

Does K meet the hypotheses?

Yes. Set

$$\kappa = \aleph_0, \qquad h(\kappa) = \beth_{\omega_2}, \qquad \lambda = \Lambda^+.$$

Under CH, K is categorical in λ, while its two endpoint models contradict categoricity at $h(\kappa)$. See 04-endpoint.tex:335 and 03-categoricity.tex:602.

Espíndola's categorical AEC definition requires accessibility, directed colimits, monomorphisms, and an iso-full, nearly full embedding into a finitely accessible category preserving directed colimits and monomorphisms (arXiv:1906.09169v7, scc3.tex:264–280). For this K:

- Countable strong submodels give accessibility; strong embeddings are monomorphisms.
- Directed colimits are concrete, with the required smoothness.
- The inclusion into the category of all structures of its finitary relational language and homomorphisms supplies the required embedding. Isomorphism invariance gives iso-fullness; coherence gives near-fullness.
- Its downward bound is exactly $\aleph_0$.

These are verified in 02-aec.tex:343, 401, 575, and 688. Internal size agrees with cardinality here (06-booleanization.tex:108–130).

Neither amalgamation nor joint embedding is part of his AEC definition or Theorem 4.1. Both fail for K (05-conclusion.tex:101). Their absence cannot exclude this example from the theorem as stated. Nor does Theorem 4.1 assume SCH or GCH.

Instantiated proof walk-through.

1. **Presentation and the presheaf point: no obstruction yet.**

   Let $\mathcal{C}_0 = K_{\aleph_0}$, using strong embeddings, and let H be the categorical model of size λ. It has countable base. Its canonical point satisfies

   $$p_H^*(y_A) = \mathrm{Hom}_K(A, H).$$

   Moreover, p_H^* preserves countable limits. Thus the objection is not a failure to obtain an $\aleph_1$-point of the **presheaf** topos (06-booleanization.tex:132–149).

   This realizes directly the evaluation construction underlying the earlier paper's classifying-topos theorem (arXiv:1906.09169v7, scc3.tex:487–504). No disputed presentation theorem is needed for this particular point.

2. **The first observation about generators: false on the literal reading.**

   Two concrete facts establish this.

   Base rigidity. If $M \leq_K N$ adds a base point, then

   $$|I^M| \leq |V_{\omega_1}| = \beth_{\omega_1}.$$

   At a new base point, every old tuple passes its test without exceptions (02-aec.tex:286–300). Well-order I^M; the diagrams for increasing finite tuples form a compatible directed system. The gluing, extensionality, and collapse argument of 03-categoricity.tex:103–194 then injects I^M into V_{ω_1}. That argument needs compatible diagrams, not colors constant by arity.

   Consequently, **every strong extension of H has exactly its base**.

   Local retraction. Suppose $H \leq_K X$, and $C \leq_K H$, $D \leq_K X$ are countable, with $C \leq_K D$ and $B^C = B^H$. There is an embedding $D \to H$ fixing C: inject the new index points into unused indices of H, retain all old fibers, and match each new tuple's fibers by the matrix argument in 03-categoricity.tex:505–599. The unchanged base makes the strong condition automatic.

   Now take any proposed generating family $\{r_i : A \to D_i\}$, and any $a : A \to H$. Enlarge H, keeping its base, to arbitrarily large cardinality. The family's advertised property supplies an amalgam with some r_i. Inside that amalgam, take countable strong submodels containing $a[A]$, B^H, and the image of D_i. Local retraction gives $D_i \to H$ factoring a.

   Thus p_H **respects every proposed generator**. The cosieves respected by a point form a Grothendieck topology, so it respects the topology they generate.

   But p_H does not respect the dense cosieve R_A below. Therefore the generated topology is strictly smaller than double negation. This proves failure of the first observation, including its assertion about the **generated** topology. Its following claim that these generators ensure Booleanness consequently fails too.

3. **"Clearly the model of size λ must be one of its points": false.**

   Choose countable $a : A \to H$ covering all of B^H, and put

   $$R_A = \{r : A \to D : r[B^A] \subsetneq B^D\}.$$

   Every countable model has a countable strong extension adding a base point, so R_A is dense.

Nevertheless, a cannot factor through any $r \in R_A$: the additional base point would have to map injectively into $B^H = a[B^A]$, colliding with an old point. Hence $p_H^*(R_A) \to p_H^*(y_A)$ is not surjective. This is precisely manuscript Proposition 7.2 (06-booleanization.tex:153–190).

This refutes the canonical model-to-point assertion. It does not classify all ordinary points of the Booleanization.

4. **Closedness does not repair this; saturation fails concretely.**

   The earlier paper defines closedness through purity of maps **out of the whole model** (arXiv:1906.09169v7, scc3.tex:617). The local-retraction argument above actually shows that H is $\aleph_1$-closed.

   Its saturation definition instead requires extension of every map from a countable model along every embedding between countable models (arXiv:1906.09169v7, scc3.tex:657–665). The map $a : A \to H$ and any base-adding $A \to A'$ witness failure of this property.

   Thus H is closed but not saturated, and its point does not Booleanize. The claim in thresholds.tex:359 identifying closed models with Booleanization points without amalgamation is false for K.

   The earlier saturation theorem explicitly assumes amalgamation (arXiv:1906.09169v7, scc3.tex:669–688); the earlier transfer uses it when passing from closedness to saturation (arXiv:1906.09169v7, scc3.tex:759–778). Removing GCH in §10 does not remove that hypothesis.

5. **Stability and the saturated slice: further missing premises.**

   In fact K is not $\aleph_0$-stable. Take countable A with one index point and fix its row representatives. For each $\alpha < \omega_1$, use the valid unary diagram with one quotient element, empty relation, and tag α. Add a new base point carrying its distinct coded color, using the construction of Lemma 7.1. These give $\aleph_1$ distinct Galois types over A: any amalgamation identifying the new points must preserve their colors against the fixed old rows. This invariant also survives chains of amalgamation identifications.

   The ingredients are 01-colors.tex:36–78 and 06-booleanization.tex:46–86.

   Therefore categoricity at λ does not supply the countable stability required when applying Lemma 2.1 to $\aleph_1$-saturation. The subsequent construction of a saturated Boolean slice containing H already lacks its designated point. I do not infer that no small saturated model exists.

6. **Sections 2–3 cannot supply the missing transfer.**

   Lemma 2.1 assumes amalgamation and stability below the saturation cardinal (arXiv:2301.13167v1, thresholds.tex:276–298). Both premises fail for the proposed application to original K.

   Theorem 3.1 requires categoricity at the lower cardinal as well as a much larger one; Theorem 3.2 requires categoricity throughout an initial interval (arXiv:2301.13167v1, thresholds.tex:328–345). Original K has neither requisite initial categoricity. The proof intends to apply them to its new saturated class, whose connection to H has failed.

   There is also unresolved parameter bookkeeping: applying Theorem 3.1 with lower categoricity cardinal κ^+ yields the displayed bound 2^{κ^+}, not automatically the 2^κ asserted in Theorem 4.1's

proof. This is an unsupported inference, not an established counterexample to tameness.

7. **The final omitting-types conclusion is false for original K.**

   The proof concludes that every sufficiently large model is κ^+-saturated, with the cutoff below $h(\kappa)$. Every tail model H above is a counterexample.

   The earlier qualitative eventual-categoricity theorem does not supply the prescribed bound: its threshold uses unspecified Hanf numbers (arXiv:1906.09169v7, scc3.tex:813–818). Qualitative eventual categoricity itself is compatible with this K.

Assessment of "same models, different morphisms."

The papers contain several distinct constructions:

- **Restriction to monomorphisms** changes nothing here: strong embeddings already are monomorphisms.
- **Substructure functorial expansion** is expressly isomorphic to the original category (arXiv:1906.09169v7, scc3.tex:720–730; arXiv:2301.13167v1, thresholds.tex:304–316). It transports the obstruction.
- **Regular Morleyization** introduces additional models until the complement axioms are restored; it is not simply the same objects with changed arrows (arXiv:1906.09169v7, scc3.tex:734–742).
- **Boolean morphisms** genuinely restrict arrows (arXiv:2301.13167v1, thresholds.tex:281). A sufficiently strong restriction can remove R_A, but can also destroy the countable presentation of H. For example, full $\aleph_1$-Boolean/elementary embeddings cannot properly extend a countable model enumerated by (a_n): they must reflect

$$\exists x \bigwedge_{n<\omega} x \neq a_n.$$

  Then H has no incoming maps from countable models, so the original countable-site construction fails.

There is an especially transparent modification illustrating the problem with the reply alone. Keep all objects but require strong embeddings to be **surjective on the base sort**. On its countable site, H extends every countable embedding by the matrix construction; its point therefore does Booleanize. Isomorphism types and the external categoricity spectrum remain unchanged.

However, this modified AEC has

$$\mathrm{LS}(K') = \aleph_1.$$

An uncountable-base object has no countable K'-submodel; conversely, retaining its entire base gives the $\aleph_1$ downward bound. Its prescribed threshold becomes exactly

$$h(\aleph_1) = \Lambda.$$

Thus this modification fixes the particular point obstruction while invalidating the proof's substitution $\kappa = \aleph_0$. It is an illustrative construction, **not a modification specified in the papers**.

I found no written modification preserving the spectrum, relevant sizes, $\mathrm{LS} = \aleph_0$, and all subsequent proof hypotheses. Arbitrary changes of arrows cannot be ruled out individually; preserving only the spectrum is insufficient.

Limits and questions for an expert.

The supplied sources do not include every cited external result, notably Makkai–Shelah's Proposition 3.3 invoked in the final footnote. I therefore cannot certify that adaptation or the full completeness/omitting-types machinery. The new rigidity, retraction, instability, and modified-category arguments above remain unformalized.

1. Does the rigidity-plus-retraction argument establish that every proposed generator is respected by p_H, hence prove failure of the first observation?
2. Precisely which modified arrows and small-object site are intended in thresholds.tex:359, and what are their LS number and presentability ranks?
3. What hypothesis replaces amalgamation in the asserted closedness-to-Booleanization passage, given the explicit closed, nonsaturated H?
4. What justifies the 2^κ bookkeeping and the final omitting-types cutoff after constructing the saturated subclass?

Claude Opus, test case

Espíndola's Theorem 4.1 tested against the CH class K (independent "informed" proofread, Opus) Read-only investigation. Sources: `[arXiv:2301.13167v1, source file thresholds.tex](https://arxiv.org/abs/2301.13167v1)` (cited as **T:line**), `[arXiv:1906.09169v7, source file scc3.tex](https://arxiv.org/abs/1906.09169v7)` (**S:line**), the manuscript TeX in `[sections](https://github.com/mathcommunity/oai-math-revised/blob/b36659d8bf74aa07719f18eecd9554bde0828873/preprints/A-CH-Obstruction-to-a-Prescribed-Categoricity-Threshold-September-24-2026/build/sections)` (cited by file:line, using the PDF numbering: Sections 2 = colors, 3 = AEC, 4 = endpoint, 5 = categoricity, 6 = conclusion, 7 = booleanization). I also read `JUDGE-REPORT.md`. Nothing here has been checked by a human.

1. Short answer

- K satisfies every hypothesis of T:354–356, read in Espíndola's own sense of AEC (S:264–282). The only escape I found needs a different value of LS (§6).
- The proof breaks in the opening paragraph of T:359. It makes two claims: (a) the topology generated by "amalgamation covers" is the double-negation topology, so the sheaf topos is Boolean; (b) the model of size λ is a point of that topos. **For K these two claims cannot both hold**, whichever reading is chosen:
  - Read so that "arbitrarily large N" includes the categorical model H: claim (a) is false. The manuscript's cosieve R_A is $\neg\neg$-dense but is not an amalgamation cover. I show below that H is a point of the generated topology, so that topology is strictly smaller than $\neg\neg$ and its sheaf topos is not Boolean.
  - Read as the full $\neg\neg$-topology on countable models: claim (b) is false (Proposition 7.2). In fact no point of that topos that preserves countable limits comes from a model of size $\geq \Lambda$.
- A later sentence in T:359 is false outright for K: "κ^+-closed models are precisely the κ^+-points of the double negation subtopos." H is λ-closed, and hence κ^+-closed, but it is not a $\neg\neg$-point. This "we do not need to use amalgamation" step is where the argument departs from scc3, which proved that characterization only under amalgamation (S:669–687).
- Later steps then fail as well:
  - the derived κ^+-AEC is said to be "categorical at κ^+ and λ", but it has no models of size $\geq \Lambda$;

  - it is said to be "categorical in a proper class";
  - "every model ... is κ^+-saturated above some $\chi < \beth_{(2^\kappa)^+}$" is also false.
- The "modified morphisms" reply does not help. Take any re-choice of morphisms on the same models that is still an AEC (isomorphism-invariant, coherent) with LS = $\aleph_0$. The R_A obstruction carries over unchanged. The only modifications that avoid it raise LS to at least $\aleph_1$. At LS = $\aleph_1$ the threshold becomes $\beth_{(2^{\aleph_1})^+} = \Lambda$, and K *is* categorical from Λ on, so there is no conflict. But that is no longer Theorem 4.1 for K, which has LS(K) = $\aleph_0$.

Overall confidence: - High: the failure at T:359 (the joint failure of (a) and (b), and the false "closed = $\neg\neg$-point" equivalence). - Medium-high: the strengthened claims, namely that H is a point of the amalgamation topology, that H is κ^+-closed, and the general morphism-change argument. - Low or undetermined: stability and the existence of saturated models of size $\aleph_1$ in K.

2. The class K (features used)

- Sorts: an index set I and a base order B (dense, without endpoints, bounded intervals countable, so $|B| \le \aleph_1$; Lemma 3.4, 02-aec.tex:167), plus row and column torsors and bits. $\|M\| = \max(|I|, |B|, \aleph_0)$ (Lemma 3.5, 02-aec.tex:190).
- Strong inclusion $M_0 \le_K M_1$*: a basic inclusion with $B^{\{M_0\}}$ convex, the same G^**, and supports preserved. In addition, at every **new** base point $b \in B^{\{M_1\} \setminus B}\{M_0\}$, every old tuple t must pass the full subsequence test T_k, with colors computed from old rows. These colors do not depend on which old rows are chosen (02-aec.tex:233–238, 286–292).
- Countable-base models with equal |I| are isomorphic (Lemma 5.9). An uncountable base forces $|I| < \Lambda$ (Proposition 5.7). So under CH the unique model H of size $\lambda \ge \Lambda$ has countable base and $|I^H| = \lambda$ (Theorem 5.10).
- LS(K) = $\aleph_0$, and internal size equals cardinality (06-booleanization.tex:114–130).

Auxiliary fact (B), which I checked myself. *If $N \in K$ has $|I^N| > \beth_{\omega_1} = |V_{\omega_1}|$, then no strong extension $N \le_K P$ adds a base point.*

Proof sketch. Suppose $b \in B^{P \setminus B}N$. At b, every tuple of I^N passes its test with row-independent colors. Fix a linear order on I^N and let D_A be the decoded diagram of the increasing enumeration of each finite $A \subseteq I^N$. The tests give $D_B{\restriction}J = D_A$ whenever $A \subseteq B$. The direct-limit and collapse argument in the proof of Lemma 5.2 (03-categoricity.tex:93–196) then goes through word for word with D_A in place of $D_{|A|}$; Remark 5.3 notes that the argument allows separate witnesses. It gives an injection $I^N \hookrightarrow Q \hookrightarrow V_{\omega_1}$, a contradiction. ■

(The same observation appears in the earlier Opus report described in JUDGE-REPORT §5. I re-derived it independently.) Under CH, $\beth_{\omega_1} < \beth_{\omega_2} < \Lambda \le \lambda$, so H cannot gain base points.

3. Walk-through of T:359, instantiated for K ($\kappa = \mathrm{LS}(K) = \aleph_0$, $\lambda \ge \Lambda$, H the model of size λ)
Step 0 (hypotheses). K is an AEC in the sense of S:264–282: - it is $\aleph_1$-accessible, with directed colimits given by unions (Lemma 3.10); - its morphisms (strong embeddings) are monomorphisms; - the forgetful functor to $\mathrm{Emb}(\tau)$ preserves directed colimits; - that functor is iso-full (isomorphism invariance, Lemma 3.9) and nearly full (coherence, Lemma 3.9).

K is categorical in $\lambda = \Lambda^+ \ge \beth_{(2^{\aleph_0})^+} = \beth_{\omega_2}$ (Theorem 5.10). **Holds.**

Step 1. "Consider the double negation subtopos of Set^{K_κ}." Here K_κ means the models of internal size $\aleph_0$ (S:337) with strong embeddings. This is exactly the manuscript's site $\mathcal{C}_0$ (06-booleanization.tex:92–106). **Well-defined.**

Step 2 (T:359, sentences 2–4). The claim is that the dense topology is generated by the families {M→N_i}, with N_i countable, such that every M→N, "for an arbitrarily large model N", amalgamates with some M→N_j. The stated reason is that such families include the excluded-middle covers built from φ and ¬φ, "which guarantees that the sheaf topos is Boolean (whence, it is the double negation subtopos)".

Instantiated for K: let J_am be the sieves with this amalgamation property, where N ranges over all of K.

- J_am is a Grothendieck topology. Pullback-stability uses LS and coherence; transitivity composes the two amalgams. I checked both.
- J_am ⊆ J_¬¬ (take N = D).
- Let A ≤_K H be countable with B^A = B^H (it exists because B^H is countable). Let R_A be the sieve of maps that add a base point (06-booleanization.tex:166).
  - R_A ∈ J_¬¬, by Lemma 7.1 and Proposition 7.2.
  - Lemma 7.1 gives ¬R_A = ∅, so R_A is precisely the cover for φ ∨ ¬φ with φ = R_A.
  - Amalgamating a : A→H with some r : A→D in R_A would yield H ≤ P with a new base point, contradicting (B).
  - So the φ/¬φ cover for φ = R_A is **not** in Espíndola's generating family. The sentence "this choice of morphisms is enough to contain all covers ... φ ... ¬φ" is **false for K**.
- More strongly, the generated topology is strictly smaller than J_¬¬, and **Sh(J_am) is not Boolean.**
  - Let J_H be the sieves that p_H sends to epimorphisms. J_H is the largest topology for which p_H is a point.
  - Every J_am-sieve S lies in J_H. Given c : A′→H, amalgamate with N = H to get H ≤ P and s : A′→D in S over A′. By (B), B^P = B^H. Choose a countable C ≤ P containing the images of D and A′ and all of B^H. Map C into H by a function g as follows:
    - g is the identity on C∩H. Over tuples from $I^{C\cap I}H$ the fibers of C, P and H coincide, because they are full G(B^H)-torsors.
    - g sends the new indices injectively to fresh indices of H.
    - On the new tuples' fibers, g is the matrix-matching torsor map from Lemma 5.9.
    - The image is a strong substructure of H: the base is not enlarged, so the strong condition is vacuous.
    - Hence c = g∘(D→C)∘s.
  - So J_am ⊆ J_H, while R_A ∉ J_H (Proposition 7.2). A dense subtopos that is Boolean must equal Sh_¬¬, so Sh(J_am) is not Boolean.
  - **Confidence: high** that the φ/¬φ claim fails. **Medium-high** for J_am ⊆ J_H: the embedding argument is mine and has not been formalized.

If instead N ranges only over countable models, J_am = J_¬¬ and Step 2 is fine. The failure then moves to Step 3.

Step 3. "Clearly the model of size λ must be one of its points." - With the ¬¬-topology this is **false for K**: by Proposition 7.2 (06-booleanization.tex:153–191), p_H does not factor through it. - Strengthening: points of Set^{$\mathcal{C}_0$} that preserve countable limits are $\aleph_1$-flat functors, which correspond to Ind_{$\aleph_1$}($\mathcal{C}_0$) ≃ K. So every κ^+-point of Sh_¬¬ is some p_M. Proposition 7.2 applies to every countable-base M of any size. Hence all κ^+-points of Sh_¬¬ have uncountable base and therefore size < Λ (Lemma 3.4 and Proposition 5.7). **Confidence: high** for H; **medium-high** for the "all κ^+-points" form, which relies on the standard identification of κ^+-points with Ind_{$\aleph_1$}.

Step 4. The model's "type" is isolated by an atom S ↣ 1. In K the subterminals of Set^{$\mathcal{C}_0$} are up-closed classes of countable models. Every countable model maps into one with |I| = $\aleph_0$, so every nonzero subterminal is ¬¬-dense and Sh_¬¬ is two-valued. Then S = 1, and **this step is vacuous**, not false.

Step 5. "There is a κ^+-saturated model of size κ^+ by stability (from categoricity at λ)", and it is a κ^+-point of the slice. **Not determined.** K lacks amalgamation, and I did not settle Galois-stability. If amalgamation holds among countable models (which looks plausible), a CH-length Fraïssé construction would give a "rich" model of size $\aleph_1$. That model would have uncountable base.

Step 6. The theory of the slice "defines a κ^+-AEC categorical at κ^+ and λ". **False for K.** By Step 3, every κ^+-point of Sh_¬¬, and so of the slice, has size $< \Lambda$. The derived class K″ therefore has no model of size λ. This holds regardless of Step 5.

Step 7 (T:359). "We do not need to use amalgamation, since κ^+-closed models are precisely the κ^+-points of the double negation subtopos". **False for K.** H is λ-closed in the sense of S:617. Proof: for f : H→N, (B) forces B^N = B^H. Any countable witness configuration in N over countable parameters from H embeds back into H over those parameters by the map g above, so f reflects the relevant existential formulas. In Rosický's categorical form: given u : A→B between countable models with f∘a = b∘u, the map t = g∘b satisfies t∘u = a. So H is κ^+-closed but not a ¬¬-point. scc3 proved "closed + amalgamation ⇒ saturated ⇔ ¬¬-point" (S:648–651, S:669–687). The missing amalgamation is exactly "H with a base-adding extension of A", which (B) forbids. **Confidence: high** that the equivalence fails; **medium-high** for the closedness proof.

Steps 8–10. - Lemma 2.1 (T:276) would make K″ an AEC. Theorems 3.1 and 3.2 (T:329, T:339) then give tameness and "categorical in a proper class of cardinals" — **false**, since K″ is bounded below Λ. Theorem 3.1's hypothesis "categorical in λ" also fails for K″. - "Categoricity everywhere above κ^+" (S:759 machinery) — **false** for K″. - The final Morley-method step, "every model of the AEC is κ^+-saturated above some $\chi < \beth_{(2^\kappa)^+}$", is **false for K**. A countable-base model exists in every cardinality $\mu \geq \aleph_0$ (Lemma 3.8), and none of them is a ¬¬-point. The contrapositive Hanf argument needed the model of size λ to be saturated, and it is not.

First failure. The first failure is in T:359, at the joint claim "Boolean (= ¬¬) **and** H is a point": - With large N, it fails at the $\varphi/\neg\varphi$ sentence. - With ¬¬ / small N, it fails at "Clearly the model of size λ must be one of its points". - The supporting sentence "κ^+-closed = ¬¬-points" is false on any reading.

Everything before T:359 that the proof cites is either not used yet or holds for K. That includes Section 3 setup T:304–326 and the scc3 results "model at a categoricity cardinal is λ-closed" (S:629) and S:489 (Theorem 4.1 of scc3, presheaf = classifying topos).

4. The "modified morphisms" reply The candidates that appear in the papers, and what happens to each:

1. **Substructure functorial expansion** (T:304; S, Section 8). The category is isomorphic to K, and density, sieves and points are invariant under equivalence of sites. **The obstruction transports.**
2. **Reg_⊥ models with homomorphisms** (T:320; S, Section 8). Espíndola asserts that Set^{K} ↪ Set$^{\{K}$r} is a *dense* subtopos (T:343, S:752). Dense subtoposes share their ¬¬-subtopos, so the ¬¬-points are the same. **No help.** (These are also not "the same models".)
3. **κ-Boolean or elementary maps** (T:281, S:790):

- L_{ω1,ω1}-elementary maps: "I is countable" is expressible in that logic, so no countable model maps into H. H is then not a colimit of countable objects, and the countable site no longer sees it. LS is no longer ℵ0.
- L_{∞,ω}-elementary maps: plausibly both a : A→H and the extensions of Lemma 7.1 are elementary, since finite-parameter back-and-forth looks routine here. If so, Proposition 7.2 goes through verbatim. (Not fully checked.)

4. **General argument.** Let $\leq'$ be any AEC structure on the same class of τ-structures with the same isomorphism classes, coherence and $LS(\leq') = \aleph_0$, whose morphisms are injective on B.
   - LS′ gives a countable A $\leq'$ H with B^A = B^H and |I^A| = ℵ0.
   - Every countable D with |I^D| = ℵ0 is isomorphic (Lemma 5.9) to some countable $D_1 \leq' M_1$, where M1 is the uncountable-base model of Theorem 4.5.
   - LS′ then gives $D_1 \subseteq D_2 \leq' M_1$ containing a further base point, and coherence′ gives $D_1 \leq' D_2$.
   - So R_A is $\leq'$-dense, and the injectivity argument of Proposition 7.2 applies unchanged.

   Hence no re-choice of morphisms consistent with "same models, still an AEC with LS = ℵ0" rescues Step 3 or Step 7.
5. Re-choices that do avoid R_A make LS at least ℵ1. One example is maps that never add base points: then M1 has no countable submodel. At LS = ℵ1 the prescribed threshold is $ב_{(2^{\aleph_1})^+} = \Lambda$, and K satisfies the conclusion. Preserving the categoricity spectrum is not enough, because the threshold depends on LS. If this is what the reply intends, it does not prove Theorem 4.1 as stated for K.

5. Could K fall outside Theorem 4.1?

- **AEC in Espíndola's sense:** yes (Step 0).
- **LS(K) = ℵ0** in Shelah's sense. Espíndola's "eventual LS number" (S:284) cannot be larger, because it is the least such bound.
- **Internal size equals cardinality**, including for countable models (06:114–130). So categoricity at λ holds in his sense too.
- **Set theory:** K needs only CH. If his proof secretly needed GCH (scc3 uses GCH in places and removes it by forcing, S:862ff), GCH implies CH, so K still applies.
- **One genuine ambiguity, in the abstract and Theorem 6.1 (T:231–236, T:382–389):**
  - The threshold there is written with κ = the accessibility index. K is ℵ1-accessible and apparently not finitely accessible.
  - With $\kappa = \aleph_1$, case 3 predicts $\chi < \Lambda$, which K does not obviously contradict. Whether uncountable-base models exist between $ב_{\omega_2}$ and Λ is not determined.
  - Theorem 4.1 itself, and T:249, explicitly use LS(K). K contradicts that statement.

Conclusion: I found no reading under which K is outside the scope of Theorem 4.1 as written.

6. What I could not determine

- Whether K is Galois-stable in ℵ0, whether amalgamation holds among countable models, and whether $Sh_{\neg\neg}(\mathcal{C}_0)$ has κ^+-points at all (Step 5).
- Full rigor of three of my own arguments: the topology-closure checks for J_am, the embedding g : C→H (the per-tuple torsor matching), and the identification of κ^+-points with $Ind_{\aleph_1}(\mathcal{C}_0)$.
- Whether Lemma 7.1's extensions and the map A→H are L_{∞,ω}-elementary.
- What exactly Espíndola means by "arbitrarily large N", by "κ^+-closed" in T:359, and by the modified category in his reply.

- I relied on, and did not re-prove, Proposition 5.7 and Theorem 4.5. JUDGE-REPORT reports checks of both; the Lean formalization covers the endpoint and tail.

7. Questions for an expert

1. In T:359, does "any M→N, for an arbitrarily large model N" range over models of size λ? If so, how does the sieve R_A (06-booleanization.tex:166) satisfy the amalgamation condition, given that H has no strong extension adding a base point (fact (B))?
2. Without amalgamation, what is the proof that κ^+-closed models are exactly the κ^+-points of Sh_¬¬(K_κ)? The model H of size λ in K appears to be λ-closed but not a ¬¬-point.
3. Which morphism class is meant in the reply? Does the modified category still have LS = $\aleph_0$? If not, is the theorem's threshold then computed from the new LS?
4. Do all κ^+-points of Sh_¬¬($\mathcal{C}_0$) for K have size $< \Lambda$ (uncountable base)? If so, what is the derived κ^+-AEC K″ "categorical at λ" supposed to be?
5. In Theorem 6.1, is κ the accessibility index or LS(K)? For AECs, which threshold is claimed: ב_{(2^{LS})$^+$} or ב_{(2^{LS$^+$})$^+$}?